كلية الهندسة بشبرا

FACULTY OF ENGINEERING AT SHOUBRA

Electrical Engineering Department
Computer Systems Engineering

Graduation Project 2025-2026

X-Fed NeuroScan

Federated Learning with Explainable AI
for Medical Image Classification

Team members

Abdelrahman Mamdouh	Abdelrahman Wahid
Goda Saber Goda	Hana Ahmed Nabhan
Mahmoud Ahmed Mahmoud	Ziad Ahmed Mohamed

Supervised by

Prof. Shaimaa Yousry
Prof. Walaa Gouda

This project is submitted as a partial fulfillment of the requirements of the degree of Bachelor of Science in Electrical Engineering (Computer Systems Engineering)

X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification

This Page is Intentionally Left Blank

Acknowledgments

We would like to express our sincere gratitude to our professors and academic supervisors for their continuous guidance, valuable feedback, and constructive insights throughout the course of this project. Their expertise and support were instrumental in shaping the direction and quality of this work.

We also wish to acknowledge our families for their unwavering encouragement, patience, and unconditional support during the development of this project. Their understanding and motivation provided a constant source of strength and inspiration.

Abstract

Medical image analysis plays a critical role in the early detection and diagnosis of various diseases. However, manual interpretation of radiographic images is time-consuming and subject to inter-observer variability. This project presents an end-to-end deep learning–based system for the automated detection of chest diseases and bone fractures from X-ray images. The proposed solution integrates multiple transfer learning models within a unified diagnostic framework, enabling efficient and scalable analysis of heterogeneous medical images.

The system architecture consists of a deep learning pipeline that includes image preprocessing, feature extraction using pre-trained convolutional neural networks, an intelligent medical image routing model, and task-specific diagnostic models. ResNet50 is employed as a core feature extractor due to its residual learning mechanism and suitability for medical image analysis, while YOLOv8 is utilized for bone fracture detection using a one-stage object detection approach. The chest disease detection task is addressed using a dedicated classification model optimized for thoracic imaging.

Preliminary experimental results demonstrate the feasibility of the proposed approach and indicate promising performance in detecting both bone fractures and chest-related conditions. The modular design of the system allows for extensibility and supports the integration of additional diagnostic tasks. This work highlights the potential of deep learning and transfer learning techniques to assist clinical decision-making and improve the efficiency of medical image diagnosis.

Table of Contents

Acknowledgments . . . I
Abstract . . . II
List of Tables . . . V
List of Figures . . . VI

1 Introduction **2**
1.1 Problem Definition . . . 2
1.2 Challenges in Medical Image Analysis . . . 4
1.3 Background and Motivation . . . 7
1.4 Suggested Solution . . . 9
1.5 Project Scope . . . 12
1.6 Project Objectives . . . 13
1.7 Project Challenges . . . 14
1.8 Summary . . . 18

2 Literature Review **21**
2.1 Medical Image Diagnosis . . . 21
2.1.1 Chest Diseases Detection . . . 22
2.1.2 Target Diseases and Their Radiological Characteristics . . . 22
2.1.3 Related Work in Chest Diseases Multi-class Classification . . . 25
2.2 Explainable AI (XAI) in Medical Image Diagnosis . . . 26
2.3 Federated Learning . . . 29
2.4 Combined Approaches . . . 31
2.5 Summary of Related Papers . . . 31
2.6 Datasets Utilized in the Project . . . 39
2.7 Similar Commercial Applications . . . 40
2.7.1 Aidoc (aiOS™) . . . 40
2.7.2 Viz.ai (Viz.ai One®) . . . 40
2.7.3 Qure.ai (qXR, qCT) . . . 41
2.7.4 Harrison.ai . . . 41
2.7.5 Lunit (Lunit INSIGHT, Lunit SCOPE) . . . 41
2.8 Summary . . . 41

3 Proposed Models **44**
3.1 Solution Architecture . . . 44
3.1.1 Deep Learning Inference Pipeline . . . 45
3.1.2 Backend Infrastructure and System Integration . . . 46
3.2 Transfer-Learning Models Used . . . 49
3.2.1 ResNet50 Model . . . 49
3.2.2 YOLOv8 Model . . . 50

3.3 Medical Image Routing Model 52
3.3.1 Motivation 52
3.3.2 Problem Statement 53
3.3.3 Model Architecture 53
3.4 Bone Fracture Detection Model 55
3.4.1 Motivation 55
3.4.2 Problem Statement 55
3.4.3 Model Architecture 56
3.5 Chest Disease Classification Model 57
3.5.1 Motivation 57
3.5.2 Problem Statement 57
3.5.3 Model Architecture 57
3.6 Summary 60

4 Results and Evaluation **63**
4.1 Evaluation Metrics 63
4.2 Results 64

References **69**

List of Tables

1.1 Key Advantages of Computer Vision in Medical Image Automation 8
1.2 Challenges of Centralized Machine Learning in Healthcare 9

2.1 Summary of related papers. 31
2.2 Summary of Datasets Used in the Project 39

List of Figures

1.1 Explainble AI in action 11
1.2 Centeralized server approach to federated learning 12

2.1 Heatmap for bone 27
2.2 Heatmap for pneumonia detection 27
2.3 FL-Based efficient framework for smart healthcare sector 29

3.1 High-level diagram of the DL inference pipeline. 45
3.2 System Architecture Overview 46
3.3 AWS S3 Integration in our system 47
3.4 Architecture of the ResNet50 illustrating the successive stages of the model. 49
3.5 Architecture of the YOLOv8 illustrating the successive stages of the model. 51
3.6 Overall architecture of the proposed medical image routing model. 54
3.7 Image preprocessing pipeline used in the attempt to train DenseNet121 on the MURA dataset. 56
3.8 High-level overview of the multi-stage chest disease classification pipeline. 58
3.9 Chest X-ray images before and after preprocessing, highlighting improved contrast and reduced noise. 58
3.10 The U-Net architecture, illustrating the encoder-decoder structure with skip connections that merge high-resolution features from the contracting path with upsampled features in the expansive path. This design is highly effective for precise medical image segmentation. 59
3.11 Application of the U-Net segmentation mask. The original preprocessed Figure 3.11a is processed to produce the final segmented lung Figure 3.11b, which serves as the input for the classification stage 59
3.12 Grad-CAM visualizations highlighting regions contributing to model predictions. 60

4.1 Training dynamics of the medical image routing model. 64
4.2 Confusion matrix of the medical image routing model. 64
4.3 Bone fracture detection model Recall-Precision curve. 65
4.4 Bone fracture detection model confusion matrix. 65
4.5 Sample output from the bone fracture detection model. 66
4.6 Classification results of the chest disease detection model. 66
4.7 Confusion matrix of the chest disease detection model. 67
4.8 Sample output from the chest disease detection model. 67
4.9 Results of successive experiments done on the MURA dataset to reach the best YOLO detection model possible. 68

CHAPTER 1
INTRODUCTION

Chapter 1 Introduction

Medical imaging plays a critical role in modern healthcare, supporting clinicians in diagnosing a wide range of conditions, from bone fractures to chest tumors and other life-threatening emergencies. As imaging technologies become more advanced and widely available, the volume of medical scans produced each day continues to grow. This rapid increase has intensified the need for efficient, reliable, and scalable diagnostic support systems that can assist healthcare professionals in interpreting complex visual data.

Recent advances in computer vision have made automated medical image analysis a promising tool for improving diagnostic accuracy and reducing clinical workload. Deep learning models, in particular, have demonstrated exceptional performance in tasks such as classification, localization, and segmentation across many imaging modalities. However, despite their capabilities, traditional centralized learning approaches often face critical limitations—especially in sensitive domains like healthcare, where data privacy, institutional policies, and regulatory constraints restrict the sharing of patient information.

To address these challenges, new paradigms in machine learning and model transparency have emerged. Federated learning enables collaborative training across multiple medical institutions without requiring direct access to patient data, offering a pathway toward more diverse and robust models while preserving privacy. Complementing this, explainable AI techniques provide insights into model decisions, helping clinicians understand why a prediction was made and increasing trust in AI-assisted diagnostics.

This project aims to integrate these three powerful components—computer vision, explainable AI, and federated learning—into a unified system designed to support medical image analysis across a variety of diseases and emergency conditions. By leveraging decentralized learning and transparent decision-making, the system seeks to improve accuracy, reliability, and interpretability while respecting the privacy and variability inherent in real-world clinical environments.

1.1 Problem Definition

Accurate diagnosis of bone fractures and chest diseases from X-ray images remains a critical challenge in modern healthcare. Many clinically important abnormalities—such as hairline fractures, early-stage pneumonia, subtle tuberculosis patterns, or small tumor-like chest masses—are visually faint, easily overlooked, or masked by overlapping anatomical structures. These challenges are intensified in high-volume clinical environments and can lead to delayed or incorrect diagnosis, negatively impacting patient outcomes.

Furthermore, deep learning models have achieved state-of-the-art performance in medical image analysis, but their clinical adoption is limited by two major obstacles: **(1) the inability to train diverse and robust models without aggregating sensitive**

patient data, and **(2) the black-box nature of deep models, which makes their predictions difficult for clinicians to trust or verify**. Hospitals typically hold isolated datasets that differ in imaging quality, patient demographics, disease prevalence, and device configurations, resulting in highly non-IID data distributions that degrade model performance when trained centrally or institutionally.

Given these challenges, the central problem addressed by this project is the following:

> **How can we build a privacy-preserving, explainable, and clinically reliable deep learning system for detecting four major chest disease classes—pneumonia, tuberculosis (TB), COVID-19, and tumor-like chest masses—using X-ray and CT images, despite highly heterogeneous data and strict privacy constraints across medical institutions?**

To contextualize this problem, the diseases and abnormalities targeted by the system are defined as follows:

- **Pneumonia:** An inflammatory lung infection characterized by alveolar consolidation, visible as opacities or infiltrates on chest X-rays.
- **Tuberculosis (TB):** A bacterial infection caused by *Mycobacterium tuberculosis*, typically presenting as lung cavities, nodular lesions, or patchy infiltrates.
- **COVID-19:** A viral infection (SARS-CoV-2) associated with ground-glass opacities, bilateral consolidations, and diffuse lung involvement on X-rays.
- **Chest masses/tumor-like abnormalities:** Suspicious opacities or lesions that may indicate malignant or benign tumors, requiring early detection for timely treatment.
- **Normal (healthy) cases:** X-rays exhibiting clear lung fields and healthy bone structures, used as the baseline class.

Addressing these conditions within a single diagnostic pipeline introduces several technical challenges:

- **Subtlety and variability of abnormalities:** Many chest pathologies present with faint or overlapping features that are difficult to detect even for experienced radiologists.
- **Data privacy and distribution constraints:** Medical data cannot be shared across hospitals due to privacy regulations, limiting model diversity and robustness.
- **Non-IID data across institutions:** Variations in imaging devices, acquisition protocols, and patient populations create distribution shifts that degrade performance.
- **Lack of interpretability:** Clinicians require transparent explanations—such as heatmaps showing fracture lines or diseased lung regions—to trust AI-assisted diagnosis.
- **Efficiency and deployability:** Models must run efficiently, with low communication costs, enabling real-world federated setups across hospitals or clinics.

Therefore, the problem tackled in this work is to design an integrated framework—*X-*

Fed NeuroScan—that combines federated learning for privacy preservation and multi-institution collaboration, with explainable AI techniques to generate clinically meaningful visual justifications for predictions, while achieving high diagnostic performance across the targeted disease categories.

1.2 Challenges in Medical Image Analysis

Medical imaging plays a fundamental role in modern healthcare, providing clinicians with valuable information for diagnosing, monitoring, and treating a wide range of diseases. Imaging modalities such as chest X-rays and Computed Tomography (CT) scans are routinely used to assess abnormalities within the thoracic region, including pneumonia, tuberculosis, pulmonary edema, and lung cancer. Despite the significant advancements in imaging technologies, the interpretation and analysis of medical images remain highly challenging tasks.

Medical image analysis traditionally relies on the expertise of radiologists and physicians who must examine large volumes of imaging data and identify subtle patterns that may indicate disease. The complexity of anatomical structures, variability among patients, and limitations of human perception can make accurate diagnosis difficult. These challenges can affect diagnostic accuracy, increase workload, and potentially delay clinical decision-making.

Complex Anatomical Structures

The human thoracic cavity contains numerous overlapping anatomical structures, including the lungs, heart, ribs, blood vessels, diaphragm, and surrounding soft tissues. In chest X-ray images, these structures are projected onto a two-dimensional plane, causing different tissues to overlap and potentially obscure important pathological findings.

The presence of overlapping structures makes it difficult for clinicians to accurately distinguish between normal anatomical variations and disease-related abnormalities. Small lesions, nodules, or early-stage infections may be hidden behind ribs or other dense structures, increasing the likelihood of missed diagnoses.

Subtle Appearance of Early-Stage Diseases

Many thoracic diseases initially manifest as subtle visual changes that are difficult to detect through routine examination. Early-stage lung cancer, for example, may appear as a small pulmonary nodule that is barely distinguishable from surrounding tissue. Similarly, mild pneumonia or early tuberculosis infections may produce only faint abnormalities within the lung fields.

Detecting these subtle findings requires significant expertise and experience. Even highly trained radiologists may encounter difficulties when abnormalities are small, poorly defined, or located in anatomically complex regions.

Variability in Disease Presentation

The same disease can appear differently across patients due to factors such as age, disease stage, genetic characteristics, and the presence of other medical conditions. Furthermore, different diseases may exhibit similar radiological appearances, making differential diagnosis challenging.

For example, pulmonary infections, inflammatory conditions, and certain types of lung cancer can produce similar opacities in chest X-rays. Likewise, benign and malignant lung nodules may share similar characteristics in CT scans. This variability increases diagnostic complexity and may lead to uncertainty during image interpretation.

Dependence on Radiologist Expertise

Accurate medical image interpretation requires extensive education, training, and clinical experience. Radiologists must possess detailed knowledge of anatomy, pathology, imaging physics, and disease progression in order to make reliable assessments.

However, diagnostic performance can vary among clinicians depending on their level of expertise and familiarity with specific conditions. Less experienced practitioners may overlook subtle findings or misinterpret complex cases. As a result, diagnostic consistency can differ between healthcare professionals and institutions.

Inter-Observer and Intra-Observer Variability

One of the well-known challenges in medical imaging is the variability in interpretation among radiologists. Different specialists may provide different assessments when evaluating the same image, a phenomenon known as inter-observer variability.

Additionally, the same radiologist may occasionally reach different conclusions when reviewing an image at different times, which is referred to as intra-observer variability. Such inconsistencies can affect diagnostic reliability and complicate treatment planning, particularly in borderline or ambiguous cases.

Large Volume of Imaging Data

Modern healthcare systems generate enormous quantities of medical images on a daily basis. Radiologists are often required to review hundreds of imaging studies during a single workday, creating significant workload pressures.

CT examinations present an additional challenge because a single scan may contain hundreds of image slices that must be carefully reviewed. Thorough examination of such large datasets is time-consuming and can contribute to fatigue, potentially increasing the risk of oversight or diagnostic errors.

Time Constraints in Clinical Practice

Healthcare professionals frequently operate under strict time constraints, particularly in busy hospitals and emergency departments. Rapid diagnosis is often essential for initiating timely treatment and improving patient outcomes.

The need to balance speed with diagnostic accuracy can be challenging, especially when dealing with complex imaging studies. Under time pressure, subtle abnormalities may be overlooked, and difficult cases may require additional consultation or follow-up examinations.

Image Quality Limitations

The diagnostic quality of medical images can be affected by several factors, including patient movement, improper positioning, low contrast, imaging noise, and equipment-related artifacts. Poor image quality may obscure important pathological findings and make interpretation more difficult.

In chest X-ray imaging, underexposure or overexposure can reduce visibility of anatomical details. In CT imaging, motion artifacts caused by breathing or patient movement can degrade image clarity. Such limitations may reduce diagnostic confidence and necessitate repeat imaging procedures.

Diagnostic Errors and Missed Findings

Despite advances in medical imaging technology, diagnostic errors remain an unavoidable challenge. Errors may occur due to perceptual limitations, cognitive biases, fatigue, distraction, or the inherently complex nature of certain cases.

Missed lung nodules, overlooked fractures, and undetected early-stage diseases represent some of the most common diagnostic challenges in thoracic imaging. Because delayed diagnosis can significantly impact patient outcomes, minimizing such errors remains a major objective within radiology.

Challenges in Lung Cancer Detection

Lung cancer detection presents particularly significant difficulties due to the diverse appearance of pulmonary nodules. Nodules may vary greatly in size, shape, texture, density, and location. Small malignant nodules can closely resemble benign lesions or normal anatomical structures.

Furthermore, certain tumors may be partially obscured by blood vessels, ribs, or surrounding tissues. Accurate identification and characterization of suspicious nodules often require careful examination across multiple CT slices and comparison with previous studies, making the diagnostic process both complex and time-intensive.

Increasing Demand for Radiological Services

The global demand for medical imaging services continues to grow due to population growth, aging demographics, and increased utilization of diagnostic imaging technologies. However, many healthcare systems face shortages of trained radiologists and imaging specialists.

This imbalance between demand and available expertise can lead to increased workloads, longer reporting times, and reduced access to timely diagnoses. The challenge is particularly pronounced in developing regions and underserved healthcare environments.

1.3 Background and Motivation

The rapid growth of medical imaging data, coupled with increased clinical workloads, has made conventional diagnostic workflows increasingly insufficient. Although automated analysis using computer vision holds considerable promise in improving diagnostic accuracy and efficiency, practical implementation is hindered by challenges such as patient privacy restrictions, limited data set diversity, and substantial computational and data transfer demands. Therefore, a thorough understanding of both the potential benefits and inherent limitations of AI in medical imaging is essential for the design of systems that are accurate, interpretable, and scalable in heterogeneous clinical environments.

Rise of Computer Vision in Automated Analysis

The rapid digitization of healthcare has led to a dramatic increase in medical imaging data, including X-ray, CT, MRI, and ultrasound scans. Radiologists now face the dual challenges of handling large patient loads and interpreting large volumes of images quickly and accurately. Computer vision has emerged as a crucial tool for addressing these demands by enhancing diagnostic efficiency and accuracy.

To understand why computer vision is now widely adopted in clinical workflows, consider the following advantages:

- **High sensitivity to subtle patterns**: AI can detect faint fractures or early-stage tumors.
- **Rapid processing**: Thousands of images can be analyzed within seconds.
- **Consistency**: Predictions remain stable regardless of fatigue or emotional state.
- **Adaptability**: Deep models can generalize to various imaging modalities.

The effectiveness of computer vision in medical imaging is reinforced by major achievements in the field. These advancements primarily arise from deep learning architectures such as CNNs and Vision Transformers. In practice, computer vision supports several core tasks:

1. Classification of abnormal vs. normal scans.

2. Localization of suspicious areas using object detection.

3. Segmentation of organs, lesions, and fracture regions.

4. Identification of patterns that deviate from healthy anatomy.

A concise summary of computer vision benefits in medical imaging is outlined in Table 1.1.

Table 1.1: Key Advantages of Computer Vision in Medical Image Automation

Computer Vision Capability	**Role in Medical Imaging**
Feature extraction	Detects subtle abnormalities beyond human perception
Automation of routine tasks	Reduces workload and radiologist fatigue
High-speed inference	Enables rapid triage in emergencies
Generalization across cases	Handles variations in anatomy and imaging conditions

Limitations of Centralized Machine Learning

Despite the promise of AI-based diagnostic systems, models trained in a centralized manner face several constraints. These limitations arise mainly from restrictions on data availability, data diversity, and practical constraints in medical settings.

Centralized learning typically suffers from:

- **Restricted data sharing**: Patient data cannot be freely exchanged due to privacy laws.
- **Lack of representation**: Single-hospital datasets fail to capture global patient diversity.
- **High storage and transfer requirements**: Medical images are large and expensive to move.
- **Data imbalance**: Rare conditions remain underrepresented, harming model performance.

These issues do not simply degrade model accuracy—they introduce significant risks into clinical AI deployment. The consequences can be better understood through the following observations:

1. Models trained in isolation often misclassify images from new machines or demographics.

2. Performance can fluctuate between institutions due to imaging protocol differences.

3. Bias can emerge when the training set fails to include certain patient groups.

Table 1.2 summarizes these core limitations in a structured manner.

Table 1.2: Challenges of Centralized Machine Learning in Healthcare

Limitation	**Impact**
Privacy restrictions	Prevent hospitals from sharing sensitive patient data
Limited dataset diversity	Reduces generalization to different populations or imaging devices
Large data transfer requirements	Increases cost and delays in model development
Imbalanced condition frequency	Degrades performance on rare diseases or subtle abnormalities

Centralized machine learning, while powerful, therefore struggles to meet the scalability, diversity, and privacy needs of real-world medical AI deployment. These challenges motivate the development of more robust learning paradigms capable of overcoming data fragmentation and institutional isolation.

1.4 Suggested Solution

The challenges identified in the medical imaging domain—including subtle abnormalities, limited dataset diversity, privacy constraints, and variability in imaging standards—necessitate a more advanced diagnostic support system. The proposed project aims to address these limitations by integrating three complementary technologies: Computer Vision, Explainable AI, and Federated Learning. Together, these components form an intelligent system capable of assisting clinicians in identifying fractures, tumors, infections, and other abnormalities more reliably and transparently.

The core idea behind the project is to design a computer vision pipeline capable of analyzing X-ray and other medical imaging modalities with high accuracy. Deep learning models will detect patterns that may be difficult for clinicians to recognize quickly, especially in emergency or high-stress situations. By augmenting image interpretation with AI-based classification and region highlighting, the system serves as a powerful second-opinion tool, reducing oversight and improving diagnostic confidence.

To ensure that the system remains trustworthy and clinically interpretable, Explainable AI techniques are incorporated to reveal how the model arrives at its predictions. This transparency allows medical professionals to understand the rationale behind the AI's outputs, compare them with clinical knowledge, and detect any mistakes or biases early.

Moreover, Federated Learning is utilized to overcome the limitations of centralized training. Instead of pooling patient data into a single server—a process that is often legally and ethically restricted—hospitals collaboratively train the model without exposing sensitive information. This approach significantly increases dataset diversity while maintaining strict compliance with privacy regulations.

In summary, the proposed solution seeks to:

- Improve diagnostic accuracy through advanced computer vision algorithms.
- Increase clinician trust via transparent and interpretable model outputs.
- Overcome data-sharing barriers by enabling collaborative, privacy-preserving training.

These three components work together to form a robust, scalable, and clinically meaningful AI-based diagnostic support framework.

Explainable AI (XAI)

Explainable Artificial Intelligence (XAI) refers to a set of methods and frameworks aimed at increasing the transparency, interpretability, and trustworthiness of machine learning models. As modern AI systems grow in complexity, their internal decision-making processes often become opaque, creating what is commonly known as the "black-box" problem. In safety-critical domains such as medicine, where clinical decisions directly affect patient outcomes, this lack of transparency is a major barrier to adoption. XAI methods attempt to address this challenge by providing human-understandable insights into how and why a model arrives at a specific prediction.

In medical imaging, explainability is particularly important due to the high stakes of diagnosis, the need for clinician trust, and the requirement for regulatory compliance. Rather than simply outputting a prediction, explainable models are capable of highlighting meaningful visual patterns, quantifying the contribution of specific features, or generating human-interpretable reasoning. These explanations serve as an additional layer of evidence that supports radiologists and clinicians during diagnostic workflows.

XAI supports clinical decision-making by:

- Identifying the precise image regions or features that contribute most strongly to the model's prediction.
- Ensuring that the model's focus aligns with clinically relevant structures rather than spurious artifacts.
- Facilitating the detection of potential model errors, such as overfitting, dataset bias, or misinterpretation of anatomical features.
- Enabling clinicians to verify, critique, or challenge AI-generated predictions using domain expertise.
- Improving trust and accountability by providing a transparent rationale that can be communicated in clinical reports.

A variety of XAI techniques are widely used in medical imaging research and practice. Visual explanation methods such as Class Activation Maps (CAM), Gradient-weighted CAM (Grad-CAM), and attention-based heatmaps highlight the spatial regions that influence a model's output. Pixel-level saliency maps capture fine-grained sensitivity to image perturbations. Beyond visual methods, some approaches provide concept-based explanations, feature attribution scores, or even natural-language descriptions that mimic clinical reasoning. Together, these techniques help bridge the gap between the sophisticated internal representations learned by deep neural networks and the need for interpretable, clinically meaningful insight within medical environments.

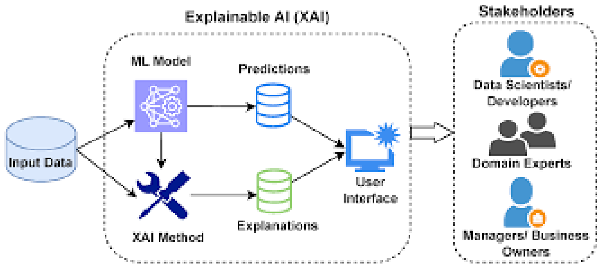


Figure 1.1: Explainble AI in action

Federated Learning

Federated Learning (FL) is a decentralized machine learning paradigm designed to enable collaborative model training without the need to share raw data between institutions. Unlike traditional centralized approaches, where data must be gathered into a single repository, federated learning allows each participating site to keep its data local while only exchanging model parameters or gradients. This architecture directly addresses major concerns in modern healthcare environments, including patient privacy, data security, legal constraints, and the challenges of aggregating large, distributed medical datasets.

In the medical field, data collected across hospitals, clinics, and imaging centers is often heterogeneous and siloed due to regulatory frameworks such as GDPR and HIPAA. Federated Learning provides a solution by ensuring that sensitive patient information never leaves its source institution. Instead, the global model is iteratively improved by aggregating updates computed locally at each site. This setup allows institutions to benefit from the collective data diversity and scale—key factors for building robust AI models—while preserving strict privacy and confidentiality guarantees.

Federated Learning benefits clinical AI development by:

- Allowing organizations to collaboratively train high-performing models without sharing **protected health information (PHI).**
- Leveraging diverse datasets from multiple geographic regions, improving the model's generalizability across patient populations.
- Reducing the risk of data breaches, since sensitive information remains stored on secure local servers.
- Supporting compliance with ethical and legal requirements governing medical data usage.
- Enabling continuous model improvement as new data becomes available in real clinical environments.

Several FL strategies have been developed to support medical imaging applications. The most widely used is *Federated Averaging (FedAvg)*, where each institution trains a local version of the model and periodically sends its parameter updates to a central aggregator. More advanced techniques—such as personalized federated learning, secure aggregation, and differential privacy—address challenges like domain shifts between institutions, communication efficiency, and protection against inference attacks. In addition, hybrid approaches integrate federated learning with techniques such as transfer learning or multimodal fusion, enabling complex models to be trained across distributed datasets while retaining strong privacy guarantees.

By enabling collaborative model development across institutions without compromising data confidentiality, Federated Learning plays a crucial role in the advancement of trustworthy and scalable AI systems in healthcare.

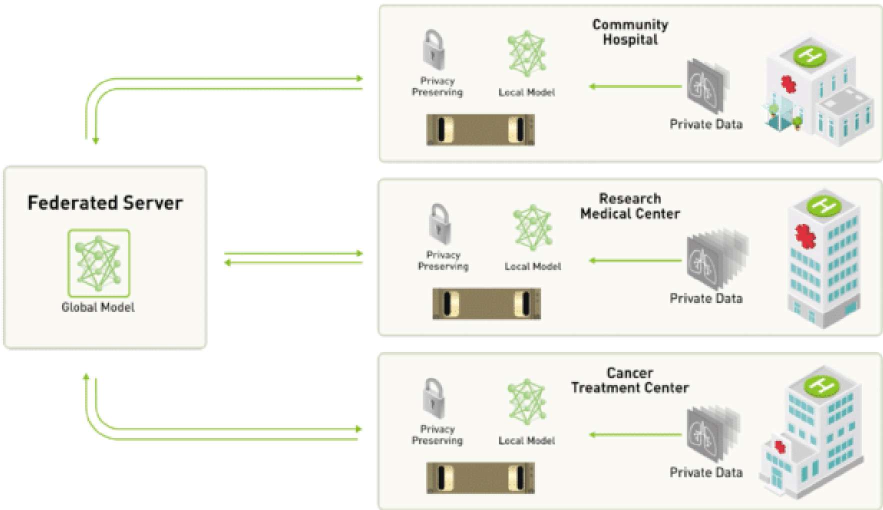


Figure 1.2: Centeralized server approach to federated learning

1.5 Project Scope

The scope of this project centers on developing a comprehensive medical image analysis system capable of diagnosing two primary categories of conditions: bone fractures and chest tumors. These conditions were selected due to their prevalence in clinical settings, their potential for severe medical consequences if missed, and the proven effectiveness of computer vision techniques in detecting abnormalities within skeletal and thoracic imaging. The system aims to support clinicians by providing fast, reliable, and interpretable diagnostic assistance.

The core of the project focuses on designing and training deep learning models that can accurately analyze X-ray images. For chest diseases and tumors, the pipeline will identify abnormal masses or opacities that may indicate malignancies or other critical conditions. The emphasis is on achieving high sensitivity and specificity while ensuring that the model remains robust across diverse imaging conditions.

To enhance usability and ensure accessibility for non-technical users, the diagnostic engine will be wrapped within a web-based application. This interface will allow clinicians, technicians, and medical staff to upload images, visualize model predictions, and interact with diagnostic outputs without requiring specialized technical knowledge. Although the interface will play an important role in facilitating real-world use, the primary focus of the project remains on the development, training, and evaluation of the underlying diagnostic models.

The system will include the following high-level functionalities:

- Automated detection of bone fractures in X-ray images.
- Identification of suspicious chest masses or tumor-like abnormalities.
- Visualization of diagnostic results and highlighted regions of interest.
- A user-friendly web interface that integrates the underlying AI engine.

The project does *not* aim to replace professional medical judgment. Instead, it seeks to provide a reliable, efficient, and interpretable tool that supports clinicians in making faster and more accurate assessments. Future extensions—such as expanding the system to additional diseases, modalities, or institutions—fall outside the current scope but remain promising directions for continued development.

1.6 Project Objectives

The objectives of this project are centered around creating a reliable, accurate, and accessible diagnostic support system for bone-fracture and chest tumor identification. The project aims to ensure that the system not only performs well in terms of machine learning metrics but also provides practical value to clinicians through usability, efficiency, and cost-effectiveness. Achieving clinical-level performance requires a well-defined set of goals that align both with technical constraints and real-world medical needs.

A primary objective is to develop a computer vision model capable of achieving high diagnostic performance. In particular, the system is designed to reach an accuracy above 90% across its targeted conditions while maintaining a recall rate that minimizes the risk of missed fractures or tumors. Since false negatives can lead to severe clinical consequences, optimizing recall is essential for ensuring the safety and reliability of the system.

In addition to predictive performance, another major objective is to deliver an intuitive and engaging web interface that enables medical personnel to interact with the system easily. The interface should allow users to upload images, view results, and interpret highlighted regions without requiring technical expertise. Ease of use is critical for facilitating adoption in real clinical workflows, where time efficiency and clarity are important.

Operational efficiency is also a key objective. The system must be designed in a manner that minimizes computational overhead and reduces ongoing costs. This includes optimizing model size, using lightweight inference strategies where possible, and ensuring that the web application remains responsive even with limited hardware resources.

To summarize, the project seeks to meet the following objectives:

- Achieve a diagnostic accuracy exceeding 90% on bone-fracture and chest tumor detection tasks.
- Maintain a high recall rate to reduce the likelihood of missed abnormalities.
- Provide a user-friendly, accessible, and engaging web interface suitable for non-technical clinical staff.
- Develop an efficient system that minimizes operational and computational costs.

Collectively, these objectives define the criteria by which the success of the project will be evaluated and guide the design decisions throughout development.

1.7 Project Challenges

Despite the remarkable progress achieved in recent years, medical image analysis remains a challenging field. The complexity of medical data, the critical nature of clinical decision-making, and the variability of imaging procedures introduce numerous obstacles that must be addressed to develop reliable and clinically applicable systems. The following subsections discuss the major challenges associated with medical image analysis, particularly in the context of chest disease detection using X-ray images and lung cancer detection using CT scans.

Limited Availability of High-Quality Medical Data

One of the most significant challenges in medical image analysis is the limited availability of large, high-quality, and well-annotated datasets. Deep learning models generally require substantial amounts of labeled data to achieve high performance and robust generalization. However, obtaining medical images with accurate annotations is often difficult and expensive.

Medical image annotation requires the expertise of trained radiologists or medical specialists who must carefully examine each image and identify pathological findings. This process is time-consuming and resource-intensive. Furthermore, patient privacy regulations and ethical considerations often restrict the sharing of medical data between healthcare institutions, limiting the availability of publicly accessible datasets.

In the context of chest X-ray analysis, some diseases may occur infrequently, leading to severe class imbalance within datasets. Similarly, for lung cancer detection from CT scans, obtaining sufficiently large collections of confirmed malignant and benign cases can be challenging. These limitations can negatively affect model training and reduce diagnostic performance.

Variability in Image Acquisition

Medical images are acquired using different imaging devices, protocols, and settings across hospitals and healthcare facilities. As a result, images may exhibit significant variations in quality, resolution, contrast, brightness, noise levels, and anatomical positioning.

Chest X-ray images obtained from different institutions may vary due to differences in machine manufacturers, patient positioning techniques, exposure settings, and image preprocessing procedures. Likewise, CT scans may differ in slice thickness, reconstruction algorithms, scanning protocols, and radiation dose levels.

Such variability can lead to distribution shifts between training and deployment environments. A model trained on data from one institution may experience performance degradation when applied to images acquired in another clinical setting. Ensuring robustness across diverse imaging conditions remains a major challenge for medical image analysis systems.

Image Noise and Artifacts

Medical images frequently contain noise and artifacts that can obscure important anatomical structures and pathological findings. These imperfections may arise from patient movement, hardware limitations, acquisition errors, or environmental factors during image capture.

In chest X-ray imaging, artifacts such as medical devices, implants, external objects, and improper positioning can interfere with disease detection. CT scans may contain motion artifacts, beam-hardening effects, metal artifacts, and reconstruction-related distortions.

The presence of noise and artifacts can significantly reduce image quality and negatively affect the performance of automated detection systems. Consequently, effective preprocessing techniques, including image enhancement, denoising, and artifact removal, are often required before analysis.

Complexity of Disease Manifestations

Many thoracic diseases exhibit highly variable visual characteristics, making automated detection particularly challenging. Diseases may present differently depending on their severity, progression stage, patient demographics, and coexisting medical conditions.

For example, pneumonia, tuberculosis, pulmonary edema, and COVID-19 may produce overlapping radiographic features in chest X-rays. Similarly, lung nodules observed in CT scans can vary considerably in size, shape, texture, density, and location. Some malignant nodules closely resemble benign abnormalities, increasing the difficulty of accurate classification.

Additionally, certain diseases may manifest as subtle changes that are difficult to detect even for experienced radiologists. Developing models capable of recognizing both obvious and subtle pathological patterns remains a critical research challenge.

Class Imbalance in Medical Datasets

Medical datasets often suffer from class imbalance, where healthy cases significantly outnumber disease-positive cases or where some diseases occur much less frequently than others. This imbalance can lead machine learning models to favor majority classes during training.

For instance, in chest X-ray datasets, normal scans may constitute a large proportion of the available data, while rare diseases may only be represented by a small number of samples. Similarly, in lung cancer screening datasets, malignant nodules may be considerably less common than benign findings.

Class imbalance can result in poor sensitivity for detecting rare but clinically important conditions. Addressing this issue often requires specialized techniques such as data augmentation, oversampling, synthetic data generation, weighted loss functions, and balanced training strategies.

Interpretability and Explainability

Healthcare professionals require transparency and trust when using artificial intelligence systems in clinical practice. Many state-of-the-art deep learning models function as black-box systems, producing predictions without providing clear explanations for their decisions.

In medical diagnosis, explainability is particularly important because clinicians must understand the reasoning behind automated recommendations before incorporating them into patient care decisions. A prediction without justification may not be sufficient in high-stakes medical environments.

To address this challenge, researchers employ explainable AI techniques such as saliency maps, Grad-CAM visualizations, attention mechanisms, and feature attribution methods. These approaches help identify image regions that contribute most strongly to model predictions, thereby improving transparency and clinical confidence.

Generalization and Robustness

A medical image analysis system must perform reliably across different patient populations, imaging devices, healthcare institutions, and geographical regions. However, models trained on specific datasets may inadvertently learn dataset-specific characteristics rather than clinically meaningful features.

This issue can lead to reduced performance when models encounter unseen data during real-world deployment. Factors such as demographic differences, variations in disease prevalence, and differences in imaging protocols can all affect model generalization.

Ensuring robustness requires diverse training datasets, rigorous validation procedures, external testing across multiple institutions, and continuous performance monitoring after deployment.

Three-Dimensional Nature of CT Data

Unlike chest X-rays, which provide two-dimensional projections, CT scans generate three-dimensional volumetric data. While this additional information improves diagnostic capabilities, it also introduces substantial computational and analytical challenges.

CT volumes may contain hundreds of slices for a single patient examination. Processing such large datasets requires considerable memory, storage capacity, and computational resources. Furthermore, accurately identifying lung nodules often requires analyzing spatial relationships across multiple slices.

Advanced techniques such as 3D convolutional neural networks, volumetric segmentation algorithms, and multi-view learning approaches are commonly employed to address these challenges. However, these methods generally demand greater computational complexity and longer training times.

Accurate Segmentation of Anatomical Structures

Segmentation is a fundamental step in many medical image analysis pipelines. It involves identifying and isolating relevant anatomical structures such as lungs, airways, lesions, tumors, and nodules.

Accurate segmentation is particularly challenging when boundaries are unclear, tissues overlap, or pathological regions exhibit irregular shapes. In chest X-ray images, overlapping anatomical structures can complicate lung segmentation. In CT scans, tumors may have poorly defined margins or be attached to surrounding tissues.

Segmentation errors can propagate through subsequent analysis stages and adversely affect disease detection and classification performance. Therefore, robust segmentation algorithms remain an important area of ongoing research.

Clinical Validation and Regulatory Requirements

Developing an accurate machine learning model is only one step toward clinical adoption. Medical image analysis systems must undergo extensive validation to demonstrate safety, reliability, and effectiveness in real-world healthcare environments.

Clinical validation typically involves testing models on large and diverse patient populations, comparing performance against expert radiologists, and assessing their impact on clinical workflows. Furthermore, healthcare technologies are subject to strict regulatory requirements and quality standards before deployment.

Meeting these regulatory and clinical validation requirements can be a lengthy and resource-intensive process, but it is essential for ensuring patient safety and maintaining trust in AI-assisted diagnostic systems.

Ethical and Privacy Considerations

Medical image analysis systems operate on highly sensitive patient information. Protecting patient privacy and ensuring compliance with healthcare regulations are critical responsibilities throughout the development lifecycle.

Issues such as data security, informed consent, data ownership, and algorithmic bias must be carefully addressed. Models trained on non-representative datasets may exhibit reduced performance for certain demographic groups, potentially leading to inequitable healthcare outcomes.

Consequently, ethical AI development requires responsible data management practices, fairness assessments, transparency, and continuous evaluation to ensure that diagnostic systems benefit all patient populations.

Computational Resource Requirements

Modern deep learning architectures used in medical image analysis often contain millions of trainable parameters and require substantial computational resources for training and inference. High-resolution chest X-ray images and volumetric CT scans further increase computational demands.

Training advanced models frequently requires powerful Graphics Processing Units (GPUs), large memory capacities, and extensive training times. These requirements may limit accessibility for smaller healthcare institutions and research groups with limited computational infrastructure.

Optimizing model efficiency while maintaining diagnostic accuracy remains an important challenge, particularly for deployment in resource-constrained clinical environments.

1.8 Summary

This chapter introduced the motivation and context behind the development of an intelligent medical image analysis system designed to support clinicians in diagnosing bone fractures and chest tumors. The rapid growth of medical imaging data, combined with the increasing complexity and subtlety of diagnostic tasks, has created a pressing need for automated tools capable of enhancing accuracy and reducing oversight. Traditional interpretation methods are challenged by factors such as faint abnormalities, variations in imaging quality, and the diverse presentation of conditions across patients and imaging devices.

To address these challenges, the chapter examined the rise of computer vision in healthcare and its effectiveness in automating image-based diagnostic tasks. While machine learning has demonstrated considerable promise in detecting complex patterns within medical images, centralized approaches face several limitations including restricted data sharing, lack of dataset diversity, and substantial computational burdens. These constraints hinder the performance and generalizability of AI models, especially in a domain where reliability is critical.

The chapter also presented the rationale for the proposed solution, which integrates advanced computer vision techniques with Explainable AI and Federated Learning. This combination aims to improve diagnostic accuracy, enhance interpretability, and overcome data privacy barriers that limit large-scale collaborative model training. By incorporating interpretability mechanisms, the system ensures that clinicians can understand and verify model decisions, while federated learning enables multi-institutional training without compromising patient confidentiality.

Furthermore, the scope and objectives of the project were clearly defined. The system aims to achieve high accuracy and recall in detecting bone fractures and chest tumors, provide a user-friendly interface suitable for non-technical users, and maintain operational efficiency to minimize computational costs. The chapter also outlined the challenges likely to arise during development, including data limitations, model robustness, explanation quality, system efficiency, and the complexities associated with implementing federated learning securely.

In summary, the introduction established the need for an intelligent, interpretable, and privacy-preserving diagnostic platform and laid the foundation for the system proposed in this project. The subsequent chapters build upon this foundation by detailing the methodologies, design choices, experiments, and evaluations that guide the development of the final system.

CHAPTER 2
LITERATURE REVIEW

Chapter 2 Literature Review

In this chapter, a literature review is provided in project-related areas. To build the framework (X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification), we need to get familiar with foundations of medical image analysis, deep learning-based classification techniques, and the challenges associated with handling sensitive clinical data. We also examine the principles and methodologies of Federated Learning (FL) as a privacy-preserving approach for multi-institutional model training, as well as Explainable AI (XAI) techniques that enhance model transparency and clinical trust. Additionally, this chapter reviews existing commercial medical imaging applications, related academic research, and publicly available datasets. Finally, we evaluate the tools, platforms, and frameworks that support the development of deep learning models, federated learning workflows, and explainable AI components.

The chapter is organized as follows:

- Section 2.1: Medical image history and challenges, with subsections on bone fractures and chest diseases.
- Section 2.2: Explainable AI (XAI) in medical image detection.
- Section 2.3: Federated Learning (FL) in medical image detection.
- Section 2.4: Combined approaches integrating detection with XAI and FL.
- Section 2.5: Summary of related papers.
- Section 2.6: Similar commercial applications.
- Section 2.7: Summary of the chapter.

2.1 Medical Image Diagnosis

Medical imaging is the cornerstone of modern diagnosis, providing non-invasive visualization of the body's interior using techniques like X-rays, Computed Tomography (CT), and Ultrasound. Due to the high volume and complexity of medical scans, interpretation is often time-consuming and subject to human variability. This process is increasingly enhanced by **Artificial Intelligence (AI)**, particularly **Computer Vision (CV)** and Deep Learning (DL) methods.

While traditional Deep Learning (DL) approaches have achieved high accuracy in radiological analysis for disease detection and bone fracture localization, their clinical deployment is significantly hindered. The primary issues stem from two critical needs: the strict requirement for **data privacy** and the **black-box nature** of AI decision-making. These barriers prevent the widespread, trustworthy adoption of powerful AI models in healthcare settings.

Consequently, the research focus has shifted toward specialized solutions: **Federated**

Learning (FL), a privacy-preserving paradigm, and **Explainable AI (XAI)**, a necessary component for model transparency. This work provides a systematic review of the state-of-the-art in these domains. It begins by examining advanced deep learning architectures, analyzing the role of FL in data collaboration, and reviewing XAI techniques.

We will first detail the related work in chest X-ray diagnosis and bone fracture detection. Subsequently, it will explore the use of XAI and FL specifically within these two application areas, identifying the gap that this research aims to bridge.

2.1.1 Chest Diseases Detection

The chest, encompassing the thoracic cavity, houses vital organs such as the lungs, heart, and major blood vessels, making it susceptible to various respiratory diseases that can be detected through imaging techniques like chest X-rays (CXR). CXR is a non-invasive, cost-effective diagnostic tool widely used for identifying abnormalities in lung tissue. We focus on four key categories for multiclassification: pneumonia, tuberculosis (TB), COVID-19, and normal (healthy) cases.

- **Pneumonia** is an inflammatory condition of the lung alveoli, often caused by bacterial, viral, or fungal infections, leading to fluid-filled sacs visible as opacities on CXR.
- **Tuberculosis** is a bacterial infection primarily caused by Mycobacterium tuberculosis, characterized by granulomas, cavities, or nodular infiltrates in the lungs, which can be chronic and contagious.
- **COVID-19**, caused by the SARS-CoV-2 virus, presents with ground-glass opacities, consolidations, and bilateral lung involvement on CXR, often progressing rapidly.
- **Normal cases** exhibit clear lung fields without pathological signs, serving as a baseline for comparison.

Accurate detection of these diseases via deep learning (DL) models on CXR images is crucial for timely intervention, especially in resource-limited settings [8] [9].

2.1.2 Target Diseases and Their Radiological Characteristics

This study focuses on three major respiratory diseases that can be detected through chest X-ray imaging: pneumonia, COVID-19, and tuberculosis. These diseases represent significant public health concerns worldwide and are among the most commonly investigated conditions in computer-aided diagnostic systems. Although all three diseases affect the respiratory system, each exhibits distinct pathological characteristics and radiographic features that can assist clinicians in diagnosis. Understanding their symptoms and appearance in chest X-ray images is essential for accurate interpretation and effective disease detection.

Pneumonia

Pneumonia is an inflammatory condition of the lungs that primarily affects the alveoli, the small air sacs responsible for gas exchange. The disease is commonly caused by bacterial, viral, or fungal infections, leading to the accumulation of fluid or pus within the affected lung tissue. Pneumonia remains one of the leading causes of respiratory-related morbidity and mortality, particularly among young children, elderly individuals, and immunocompromised patients.

Common symptoms of pneumonia include:

- Persistent cough, often accompanied by mucus production.
- Fever and chills.
- Shortness of breath.
- Chest pain, especially during breathing or coughing.
- Fatigue and general weakness.
- Rapid breathing in severe cases.

In chest X-ray images, pneumonia typically appears as areas of increased opacity, commonly referred to as consolidations or infiltrates. These opacities occur due to the accumulation of inflammatory fluids within the alveoli, reducing the normal air content of the lungs. Depending on the severity and type of infection, the abnormalities may be localized to a single lobe (lobar pneumonia) or spread across multiple regions of the lungs. The affected areas generally appear whiter than healthy lung tissue, which normally appears dark due to the presence of air.

COVID-19

Coronavirus Disease 2019 (COVID-19) is a highly contagious respiratory illness caused by the SARS-CoV-2 virus. Since its emergence, COVID-19 has posed substantial challenges to healthcare systems worldwide due to its rapid transmission and potentially severe respiratory complications.

The clinical presentation of COVID-19 varies considerably among patients, ranging from asymptomatic infections to severe respiratory failure. Common symptoms include:

- Fever.
- Dry cough.
- Fatigue.
- Shortness of breath.
- Loss of taste or smell.
- Sore throat.
- Muscle and body aches.
- Headache.

Chest X-ray findings associated with COVID-19 often include bilateral lung abnormalities, particularly in the peripheral and lower lung regions. The disease commonly manifests as patchy or diffuse opacities that may affect both lungs simultaneously. As the infection progresses, these opacities may increase in size and density, resulting in widespread pulmonary involvement. Severe cases can demonstrate extensive bilateral infiltrates that significantly reduce the visibility of normal lung structures. Although chest X-rays are useful for evaluating disease progression, early-stage COVID-19 infections may not always produce clearly visible abnormalities.

Tuberculosis

Tuberculosis (TB) is a chronic infectious disease caused by the bacterium *Mycobacterium tuberculosis*. The disease primarily affects the lungs, although it can also spread to other organs. Tuberculosis remains a major global health concern, particularly in regions with limited healthcare resources.

The symptoms of pulmonary tuberculosis often develop gradually and may include:

- Persistent cough lasting several weeks.
- Coughing up blood or blood-stained sputum.
- Chest pain.
- Fever.
- Night sweats.
- Unexplained weight loss.
- Fatigue and weakness.
- Loss of appetite.

Tuberculosis exhibits a wide range of radiographic appearances depending on the stage and severity of infection. In chest X-ray images, the disease frequently affects the upper lung regions and may appear as patchy infiltrates, nodular lesions, or areas of consolidation. Advanced cases can produce cavitary lesions, which are hollow spaces formed by tissue destruction within the lungs. Fibrosis and calcified granulomas may also be observed in chronic or healed infections. The diverse presentation of tuberculosis can sometimes complicate diagnosis, as certain imaging features overlap with those of other pulmonary diseases.

Comparative Radiological Characteristics

Although pneumonia, COVID-19, and tuberculosis all affect the respiratory system and produce abnormalities in chest X-ray images, their radiographic patterns often differ. Pneumonia commonly presents as localized consolidations caused by fluid-filled alveoli, whereas COVID-19 typically produces bilateral and diffuse opacities that frequently involve the peripheral lung regions. Tuberculosis often demonstrates upper-lung involvement, nodular infiltrates, and cavitary lesions, particularly in advanced stages of the disease. Recognizing these characteristic imaging patterns is crucial for differentiating between conditions and supporting accurate clinical diagnosis.

2.1.3 Related Work in Chest Diseases Multi-class Classification

Interpreting Chest X-rays (CXRs) is vital for diagnosing various thoracic diseases like pneumonia and tuberculosis. Due to the subtlety of findings and high image volume, **Deep Learning (DL)** models have been widely adopted to automate diagnosis. These models, trained on large datasets, primarily focus on **multi classification** of different chest diseases. The following research examples showcase the success of various **Convolutional Neural Network (CNN)** architectures in achieving high accuracy in CXR analysis.

- The researchers Develop an automated Deep Neural Network (CNN) to classify Pneumonia, COVID-19, Tuberculosis, and No-findings from chest X-rays, addressing limitations in individual disease detection and small/biased datasets in prior studies. they Created a custom CNN architecture with 5 convolutional blocks (16-256 channels, ReLU activation, max-pooling), trained on split data (80% training, 20% testing). They Addressed heterogeneous data sources by preprocessing images (resizing to 300x300, denoising, normalization, filtering) to create a coherent dataset of 21,581 images from multiple Kaggle sources they achieve accuracy 98.72% (using validation metrics) , Outperformed SOTA schemes like Venkataramana et al. (96.6%) and Bhandari et al. (94.54%), The study proved that a large diverse dataset is essential to achieve high multiclass accuracy and prevent overfitting, though class imbalance slightly reduced COVID-19 performance (96.27% recall) [10].

- The researchers Develop a custom CNN model with soft attention to classify chest X-ray images into normal, COVID-19, pneumonia, and tuberculosis, while analyzing misclassifications through handcrafted features and statistical methods to improve clinical interpretability and diagnostic reliability. , Utilized transfer learning with pre-trained CNN models (VGG-19, ResNet-50, EfficientNet, DenseNet), trained on a balanced dataset split into training (70%), validation (15%), and testing (15%) sets. They Preprocessed images via normalization, data augmentation (rotation, flipping, zooming), and resizing; applied SMOTE at the feature level to balance the originally imbalanced dataset from heterogeneous sources Best Modelis ResNet-50 with Accuracy: 97% Compared to VGG-19 (95%), DenseNet (91%), EfficientNet (94%) [11].

- This study proposed a multi-level classification model using CNNs to differentiate COVID-19 from Tuberculosis (TB) and other Pneumonias based on CXR images. The primary challenge was addressing the severe data imbalance, specifically the scarcity of COVID-19 samples. The novelty involved combining data augmentation with the SMOTE technique to artificially balance the minority class. The system employed a two-stage process: first, classifying TB versus general Pneumonia, and second, sub-classifying the Pneumonia cases into COVID-19 and viral/bacterial types. This multi-level approach was designed to significantly reduce misdiagnosis between these similar lung diseases. The model achieved a high accuracy of 97.4% for the TB/Pneumonia stage. The second stage achieved 88% accuracy for bacterial/viral/COVID-19 classifications, demonstrating an overall 8–10% improvement over existing methods [12].

- Additionally, a unified DL framework like CXR-MultiTaskNet used ResNet50 for joint disease localization and classification, incorporating thoracic diseases like pneumonia and TB, with an F1-score of 96.5% through shared feature extraction [13].

- Another deep-learning pipeline classified chest X-ray images into normal, pneumonia, TB, and COVID-19 using a sequential CNN approach. Preprocessing steps such as grayscale conversion and normalization improved robustness, while transfer learning enhanced performance on limited data. VGG-16 achieved 95.9%, DenseNet-161 reached 98.9% for distinguishing COVID-19 from pneumonia, and ResNet-18 obtained 76% for COVID-19 severity classification. The study demonstrates the effectiveness of multi-stage CNN models for automated COVID-19 screening from CXR images[14].

- A knowledge-distillation deep-learning model classified CXR images into COVID-19, pneumonia, TB, and normal using the RDD4 dataset. A deep CNN teacher network transferred its knowledge to a lightweight student model built in Keras/TensorFlow, preserving accuracy while reducing complexity. Data imbalance was addressed through augmentation. The distilled student model achieved 96.08% accuracy, with high precision and recall, showing strong suitability for real-time, low-resource deployment[15].

- They develop a custom CNN model with soft attention to classify imbalanced chest X-ray images into normal, COVID-19, pneumonia, and tuberculosis, the approach Designed COV-X-net19, a 16-layer CNN architecture (4 conv layers, 6 max-pooling, soft attention, dropout, dense with SoftMax), trained on split data (70% training, 10% validation, 20% testing) with ablation studies for hyperparameter optimization (ReLU, Nadam optimizer, LR=0.001, batch=32) , the technique used preprocessed images via text removal (OCR+inpaint), morphological opening (3x3 kernel), CLAHE (ClipLimit=5.0, TileGridSize=3x3), histogram equalization. Best Model: COV-X-net19 with accuracy: 95.18% Compared to 8 trans fer learning models (e.g., MobileNet at 82.96%, VGG16 at 62.46%), base CNN (89.96%), and SOTA studies with similar multi-class setups (e.g., accuracies below 95% in imbalanced scenarios) [16].

- A CNN-based system classified chest X-ray images into COVID-19, pneumonia, TB, and normal using over 21,000 images from three public datasets. The model used five convolutional blocks with preprocessing steps such as resizing, denoising, and normalization to handle heterogeneous sources. It achieved a high validation accuracy of 98.72%, with strong per-class performance across all categories. The study shows that a carefully designed CNN can effectively support joint diagnosis of major chest diseases from CXR images [17].

2.2 Explainable AI (XAI) in Medical Image Diagnosis

Explainable AI (XAI) refers to techniques that make the decision-making process of AI models transparent and interpretable, addressing the "black-box" nature of DL models by providing insights into why a prediction was made. In this project, XAI is integrated using methods like Gradient-weighted Class Activation Mapping (GradCAM), which generates heatmaps highlighting regions in medical images (e.g., fracture lines in bone X-rays or opacities in CXR) that influenced the model's classification. This linkage enhances trust in diagnostics for both bone fractures and chest diseases, allowing clinicians to verify AI outputs against radiological features, reducing misdiagnosis risks, and facilitating model debugging.

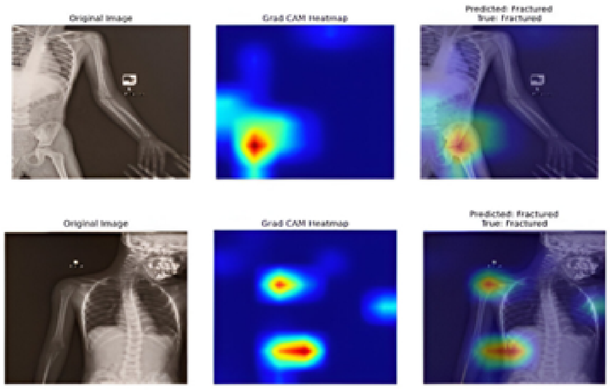


Figure 2.1: Heatmap for bone

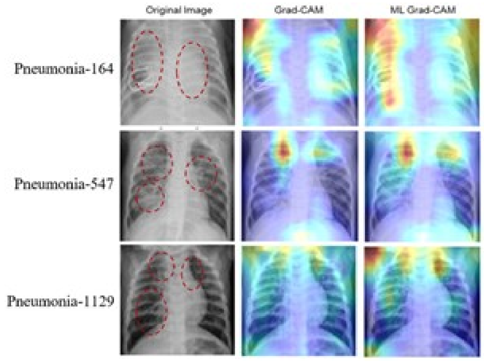


Figure 2.2: Heatmap for pneumonia detection

Related Papers on XAI For Bone Fracture Detection

- Explainable AI has been increasingly integrated into medical imaging to ensure transparency One study enhanced wrist fracture detection and classification by combining deep learning with XAI techniques, using saliency maps to highlight image regions that influenced predictions, thereby improving interpretability for clinicians. This work was presented in Enhancing Wrist Fracture Detection and Classification through Deep Learning and XAI (IEEE, 2024) [18].

- Another contribution focused on radiology, where machine learning models for bone fracture detection were augmented with XAI methods to visualize decision making processes. By providing interpretable outputs, the study bridged the gap between algorithmic accuracy and clinical usability. This was detailed in Enhancing Bone Fracture Detection in Radiology: A Machine Learning and Explainable AI Approach (IEEE, 2024) [19].

- Beyond fracture detection, interpretable machine learning has been applied to osteoporosis risk prediction. Comparative evaluations of predictive models were enriched with XAI analysis, allowing clinicians to understand how risk factors contributed to outcomes. This was reported in Interpretable Machine Learning for Osteoporosis Risk: A Comparative Evaluation of Predictive Models and XAI Analysis (IEEE, 2024) [20].

- Finally, advanced computational models have been proposed for predicting bone fracture healing and structural repair. XAI methods were employed to clarify how features influenced healing trajectories, supporting decision making in treatment planning. This was explored in XAI based Advanced Computational Models for Predictive Analysis of Bone Fracture and Structural Repair Healing Prediction

(IEEE, 2024) [21].

- This study presents an Explainable AI system for humerus fracture detection utilizing deep learning models trained on 1,266 X-ray images from the MURA dataset. The researchers found that an ensemble of DenseNet architectures achieved a superior accuracy of 85%, significantly outperforming standalone models like CNNs and VGGs. By integrating Saliency Maps and GRAD-CAM, the system visualizes specific fracture locations to ensure clinical interpretability and build trust with medical practitioners. This approach successfully combines high-performance automation with the necessary transparency for reliable orthopedic diagnosis [22].

- This study introduces a transparent AI framework for hand fracture detection using the YOLOv5m model trained on over 9,000 X-ray images. Optimized with specific preprocessing, the system achieved a superior 95.87% accuracy, significantly outperforming architectures like Faster R-CNN in both speed and precision. The integration of Grad-CAM provides visual heatmaps with over 85% localization accuracy, verifying that the model focuses on actual fracture lines rather than noise. This approach delivers a high-speed, interpretable diagnostic tool capable of enhancing clinical decision-making and trust [23].

Related Papers on XAI For Chest Disease Multi-class Classification

- This research introduces an Explainable AI (XAI) framework for the automatic detection, interpretation, and explanation of pulmonary diseases from chest radiographs (CXRs). The framework uses a transfer learning approach with a fine-tuned ResNet50 Convolutional Neural Network (CNN) trained on the COVID-CT and COVIDNet datasets. The Local Interpretable Model-agnostic Explanations (LIME) technique is employed to visually highlight the key regions in the CXR images that contribute most to the prediction. The developed system demonstrated strong performance, achieving high classification accuracy, reaching up to 97% on the evaluated datasets. Validation confirmed that the model accurately focuses on clinically important features, thus assisting radiologists in early diagnosis and clinical decision-making[24].

- They develop XAI-TRANS, using inception-based transfer learning to classify CXR images for COVID-19, bacterial/viral pneumonia, and tuberculosis, addressing overlapping radiological features and black-box AI limitations by integrating improved U-Net segmentation and XAI tools the approach used is Utilized transfer learning with four pre-trained models (ResNet50, VGG19, VGG16, Inceptionv3), combined with an improved U-Net for lung segmentation to extract features, followed by XAI-TRANS for classification and explanation using Grad-CAM visualizations Technique Preprocessed images via resizing to 256*256 and normalization between 0 and 1fer learning models (e.g., MobileNet at 82.96%, VGG16 at 62.46%), base CNN (89.96%), and SOTA studies with similar multi-class setups (e.g., accuracies below 95% in imbalanced scenarios) the Best Model is XAI-TRANS (with and Grad-CAM) with accuracy: 97% (overall classification accuracy) and 98% precision, outperforming traditional DL methods by 4.75% through XAI integration [25].

- A survey reviewed over 200 papers on XAI in DL-based medical image analysis, emphasizing visual explanations for chest radiography to build clinician confidence[26].

- Additionally, an XAI-integrated DL framework for multiclass lung disease classification, targeting five classes: viral pneumonia, bacterial pneumonia, COVID-19, tuberculosis, and normal. It evaluates various DL architectures, including CNNs, hybrids, ensembles, transformers, and Big Transfer, with hyperparameter tuning and stratified 5-fold cross-validation to ensure robust performance. The Xception model, enhanced with transfer learning, emerges as the top performer, achieving 96.21% accuracy by focusing on discriminative features in lung region[27].

- They built a lightweight CNN (5 convolutional layers) to classify chest X-rays into four classes: COVID-19, Pneumonia, TB, Normal. They applied Grad-CAM, LIME, and SHAP to generate visual explanations of model decisions, validating them with radiologists. The dataset was 7,132 public CXR images from three sources, with class imbalance handled by stratified batching. The model achieved a 10-fold cross-validation average test accuracy of 94.31% ± 1.01%, and they reported strong precision, recall, and F1[28].

2.3 Federated Learning

Federated Learning (FL) is a decentralized ML approach where models are trained collaboratively across multiple devices or institutions without sharing raw data, preserving privacy through local updates aggregated centrally. In medical imaging like bone X-rays and CXR classification, FL is used to train on distributed datasets from hospitals, mitigating data silos and complying with regulations like GDPR or HIPAA. It involves clients training on local data, sending model weights to a server for aggregation (e.g., via FedAvg), and is quickly implemented for real-time applications, reducing communication overhead while handling heterogeneous data.

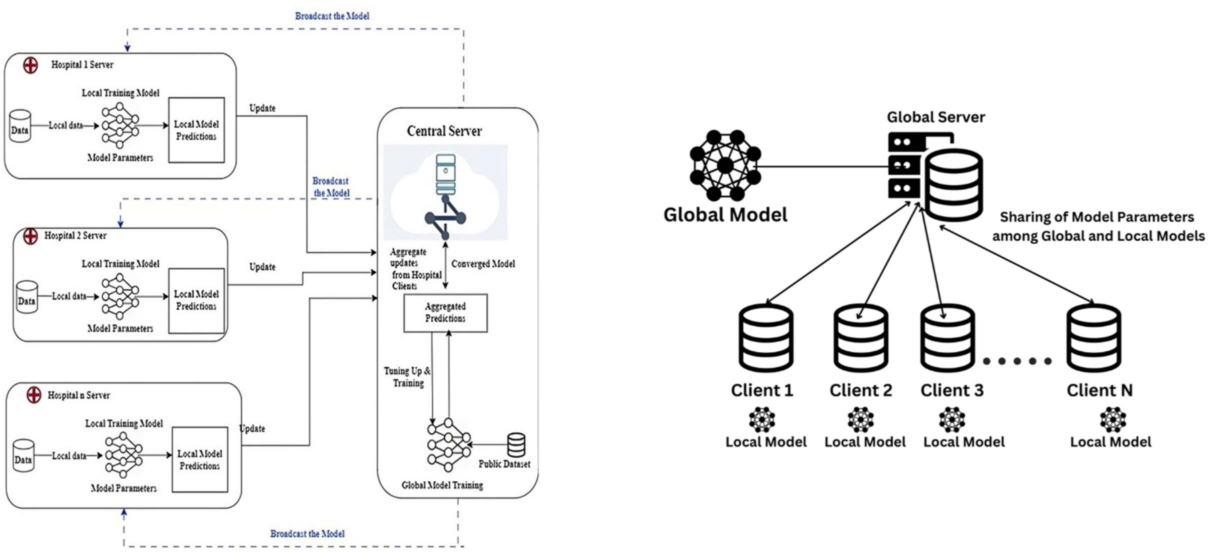


Figure 2.3: FL-Based efficient framework for smart healthcare sector

Related Papers on FL For Chest X-ray Disease Multi-class Classification

- This study addresses the primary challenge in applying Federated Learning to chest X-ray analysis, which is the significant non-IID data distribution between adult and pediatric cohorts from different institutions. Analyzing over 400,000 radiographs, the research showed that conventional FL struggles to maintain consistent performance, particularly with the challenging pediatric dataset. The authors proposed enhancing FL by integrating general-purpose Self-Supervised Learning representations, using the DINOv2 model as a robust initialization.vThe new SSL-equipped framework proved highly effective, achieving an improvement exceeding 20% in the AUC metric for pneumonia diagnosis in the difficult pediatric cases. This validates that readily available SSL models provide a practical and scalable solution for enabling FL in diverse healthcare settings while preserving data privacy[33].

- This study focuses on using federated learning (FL) for the accurate and privacy-preserving diagnosis of pediatric pneumonia via chest X-ray classification. Traditional machine learning approaches face significant challenges related to centralizing sensitive medical data, which risks patient privacy and security. FL addresses this by enabling collaborative model training across different institutions while ensuring the raw patient data remains local and secure. The researchers specifically tackled the common problem of data heterogeneity, which typically compromises classification accuracy in real-world clinical settings. They proposed an improved FL method utilizing "two-end control variables," modifying the loss function on the client side and adding momentum on the server. This novel approach successfully enhanced performance, achieving an average accuracy of 98% and demonstrating an approximate 2% improvement over existing federated learning algorithms. [33].

- Artificial Intelligence for lung disease detection is limited by major privacy concerns due to the need to centralize sensitive patient data (X-ray and CT scans). To overcome this, the paper proposes a decentralized Federated Transfer Learning (FTL) system, which enables effective collaborative model training without ever pooling raw data. This approach leverages federated learning's privacy features and the efficiency of transfer learning to train models on limited local datasets. The study evaluated four methods using the ResNet-50 model, confirming FTL as the most effective method for high accuracy. Specifically, the resulting accuracies were: Centralized (87.95%), Transfer (88.04%), Federated (87.55%), and FTL (89.96%) [34].

- A federated learning framework distinguished COVID 19 from four other chest diseases (lung cancer, TB, pneumothorax, pneumonia) plus normal using chest X rays from multiple hospitals. They used three pre trained models (DenseNet 169, VGG 16, VGG 19) and applied SMOTE to mitigate class imbalance. The aggregated system (using DenseNet 169) achieved 98.45% accuracy, Grad CAM was used for model explainability, and their decision making client selection strategy reduced communication overhead significantly [35].

2.4 Combined Approaches: Medical Image Diagnosis with XAI and Federated Learning

Integrating XAI and FL in medical image diagnosis combines privacy with interpretability, enabling collaborative training on bone X-rays and CXR while explaining predictions, with potential for broader medical domains. This section details key papers that merge these elements.

Related Papers on Combined Approaches for Chest Diseases Detection with XAI and Federated Learning

- A Federated Learning Based Real Time Ensemble Model with Explainable AI Framework for an Efficient Diagnosis of Lung Diseases. It is FLEM-XAI, a privacy-preserving FL framework with XAI for real-time lung disease diagnosis from CXR, using FedAvg for decentralized training across hospitals without data sharing. It ensembles pre-trained models (InceptionV3, Conv2D, VGG16, ResNet-50) trained locally, aggregated centrally, and incorporates XAI via SHAP for feature importance and GradCAM for heatmaps highlighting affected lung areas. Results show ResNet-50 achieving 95.5% overall accuracy (97.72% COVID-19, 95.42% pneumonia, 93.17% TB), outperforming centralized models in bandwidth efficiency and maintaining convergence [37].
- This study introduces FLPneXAINet, a framework that securely predicts pneumonia from Chest X-ray (CXR) images by integrating Federated Learning (FL), Deep Learning (DL), and Explainable AI (XAI) to overcome data privacy issues. The method utilized an 8,402-image Kaggle dataset, which was augmented using a CycleGAN network, and then features were extracted via pre-trained DL models and four Ensemble DL (EDL) models. Feature optimization was performed using RFE, ANOVA, and RF to select the most relevant features, which were then classified by various machine learning models in an FL environment. The EDL network achieved the best results in the FL environment, with a high accuracy of 97.61%, and was validated using XAI techniques (LIME and Grad-CAM). The full performance metrics were: F1 score 98.36%, recall 98.13%, and precision 98.59%, offering a robust solution for healthcare professionals [38].
- A secure lung cancer prediction used mapreduce blockchain FL with XAI, ensuring interpretability in federated settings [39].

2.5 Summary of Related Papers

Table 2.1: Summary of related papers.

Focus Area	Sub-category	Model	Dataset	Result	Ref. Num.
Bones	General	Custom CNN and transfer learning	**FracAtlas**	The custom CNN achieved superior performance with 95.96% accuracy, significantly outperforming the transfer learning models (EfficientNetB0, MobileNetV2, ResNet50) which only managed between 65% and 67% accuracy.	[1]
Bones	General	YOLOv11 was developed and benchmarked against **Faster R-CNN** and SSD.	—	YOLOv11 achieved superior performance with 96.8% mAP and 95.14% accuracy, significantly outperforming Faster R-CNN (87.5% mAP) and SSD (82.9% mAP) in both real-time fracture detection and localization.	[1]
Bones	General	YOLOv11 was developed and benchmarked against Faster R-CNN and SSD.	—	YOLOv11 achieved superior performance with 96.8% mAP and 95.14% accuracy, significantly outperforming Faster R-CNN (87.5% mAP) and SSD (82.9% mAP) in both real-time fracture detection and localization.	[1]
Bones	General	EfficientNet-B6	FracAtlas	The EfficientNet-B6 model achieved a 96.83% testing accuracy	[3]
Bones	General	Custom CNN	——	The model achieved a classification accuracy of 92.44%	[4]
Bones	General	SVM, k-NN, Naïve Bayes	———	Support Vector Machine (SVM) classifier achieved the highest accuracy, with a classification rate of 98%.	[5]
Bones	General	Custom CNN		Accuracy: 89.865% (Approx. 90%), Specificity: 89.90%	[6]
Bones	General	Custom CNN	MURA dataset	Testing Accuracy: 99.79% (at epoch 13).	[7]

Bones	XAI	MobileNetV3 style convolutional blocks with self-attention layers.	NEU Surface Defect Database, MVTec AD (Anomaly Detection), Severstal Steel Defect Dataset	proposed EdgeViT-Slim achieved 96.0% accuracy, 0.953 F1 score, and 85 FPS on industrial defect datasets, outperforming baselines with far fewer parameters	[18]
Bones	XAI	ResNet-50, Faster R-CNN, and Mask R-CNN with Grad-CAM explainability	MURA	achieving 83% classification accuracy (ResNet-50)	[19]
Bones	XAI	Random Forest, SVM, and KNN	MURA	Random Forest classifier achieved the best performance with 95% accuracy, 93% precision, and 0.96 F1-score, outperforming SVM (91%) and KNN (88%)	[20]
Bones	XAI	Logistic Regression, Random Forest, XGBoost, AdaBoost, LightGBM, and Gradient Boosting	custom multi-center clinical osteoporosis dataset (1,250 records)	ensemble model achieved the best performance with 90.82% accuracy, 100% precision, 81.63% recall, and 89.99% F1-score	[21]

Bones	XAI	(CNN, VGG, DenseNet) and found that Ensemble-2 (combining DenseNet121 and DenseNet169)	MURA dataset	The Ensemble-2 model achieved the highest accuracy of 85% and an F1 score of 0.79. Additionally, the integration of Explainable AI (Grad-CAM and Saliency Maps)	[22]
Bones	XAI	YOLO, Faster R-CNN,	——-	YOLOv5m (Best Performer) Achieved a Pointing Game Accuracy of > 85%, meaning Grad-CAM heatmaps consistently aligned with the ground truth fracture locations.	[23]
Bones	FL	DenseNet161		Federated Learning (FedAvg): The 5-node FedAvg model achieved the best performance with 91% accuracy. Centralized Training: The centralized DenseNet161 model achieved 89.6% accuracy.	[29]
Bone	FL	Local ML Models (Client-side), Federated Learning Frameworks (Server-side)	–	Highest accuracy of 94.4% (FL+VGG16 with augmentation) for binary classification.	[30]
Bones	FL	EfficientNet V2 (implemented within an Adaptive Federated Learning framework)	—	federated methods lagged slightly behind the 96–98% accuracy achieving near-centralized performance of up to 98.6% using EfficientNetV2	[31]

Bones	Combined	SemFedXAI (a framework combining Semantic Web technologies and Federated Learning	——	The SemFedXAI framework achieved the highest predictive accuracy at 85.3%, outperforming standard Federated Learning (73.5%) and other baselines. It also demonstrated superior explainability, scoring 0.85 in feature identification accuracy	[36]
Chest	General	Custom CNN	Kaggle & IEEE	The model achieved 98.72% overall accuracy across all classes. Recall per class: Pneumonia 99.66%, No-findings 99.35%, Tuberculosis 98.10%, COVID-19 96.27%.	[10]
Chest	General		Kaggle	Accuracy: 97% (using Resnet-50) VGG-19 (95%), DenseNet (91%), EfficientNet (94%)	[11]
Chest	General	Custom CNN	Kaggle	TB vs Pneumonia classification: 97.4% accuracy; COVID-19 vs viral/bacterial pneumonia vs (other pneumonia types) classification: 88.0% accuracy	[12]
Chest	General	Custom CNN	VinDr-CXR	Classification Accuracy = 96.8% (from micro-F1 ≈ 0.968) Localization Accuracy ≈ 85% (IoU ≈ 0.851)	[13]
Chest	General	VGG16, DenseNet-161 and ResNet-18	Custom clinical dataset collected and labeled	Normal/Pneumonia/TB/ COVID-19 classification (via VGG-16): 95.9% . Normal/Pneumonia/COVID-19 classification (via DenseNet-161): 98.9% . COVID-19 severity classification (via ResNet-18): 76.0% .	[14]
Chest	General	Custom CNN	Kaggle	97% average accuracy (with precision ≈ 94% and recall ≈ 97%) for the full classification task.	[15]

Chest	General	Custom CNN and VGG-16	Kaggle & public dataset from Github	The COV-X-net19 achieves 95.18% , MobileNet : 82.96%, VGG16 : 62.46% and base CNN : 89.96% .	[16]
Chest	General	DenseNet201, ResNet152V2	IEEE	The model achieved superior performance with a test accuracy of 98.87%, outperforming the individual base models (DenseNet201 and ResNet152V2) and demonstrating robust multi-class classification capabilities.	[17]
Chest	XAI	Explainable Framework based on DenseNet-201	–	The proposed explainable DenseNet-201 model achieved a high classification accuracy of 98.08%, while also providing visual explanations (Grad-CAM) to build trust and interpretability for clinical diagnosis.	[24]
Chest	XAI	CNN and Vision Transformer	Kaggle	XAI-TRANS achieves accuracy: 97% and 98% precision	[25]
Chest	XAI	Review of various Deep Learning models and XAI techniques (e.g., Grad-CAM, LIME)	Survey of medical image analysis studies		[26]
Chest	XAI	Custom CNN with Grad-CAM and LIME	–	The proposed Custom CNN model achieved a high classification accuracy of 98.41%, with XAI techniques successfully highlighting the critical image regions used for diagnosis, thereby improving model interpretability.	[27]

Chest	XAI	Ensemble of DenseNet201, Inception-ResNetV2, Xception with Grad-CAM++	Kaggle & IEEE	The proposed deep learning ensemble achieved a superior average classification accuracy of 98.97%, with XAI visualizations providing clinically plausible explanations for the model's decisions across all disease classes.	[28]
Chest	FL	Federated Learning (FedAvg) with a ResNet-based backbone and a novel age de-biasing loss function.	Multi-institutional chest X-ray datasets	The proposed de-biased FL model achieved an AUC of 89.7%, outperforming the standard FL model (AUC of 86.2%) and demonstrating significantly improved fairness and accuracy across different age groups.	[32]
Chest	FL	Federated Averaging (FedAvg) algorithm with a DenseNet-121 architecture as the core CNN model.	Chest X-ray images from multiple clients	The federated DenseNet-121 model achieved a test accuracy of 96.7% and an F1-score of 95.8%, matching the performance of a model trained on centralized data while preserving privacy.	[33]
Chest	FL	Federated Transfer Learning using a pre-trained ResNet-50 model, fine-tuned collaboratively across sites.	Collection from Kaggle	The federated transfer learning model achieved a classification accuracy of 96.3% for detecting various lung diseases, significantly outperforming models trained on single, isolated datasets.	[34]

Chest	FL	A custom Federated Learning framework using a specialized Composite-DenseNet	–	The DMFL_Net framework achieved a high overall classification accuracy of 98.21% and a precision of 98.14%, demonstrating superior performance for multi-disease classification in a federated setting.	[35]
Chest	Combined	A Federated Learning-based Real-Time Ensemble Model combining multiple CNNs (VGG16, ResNet) with XAI techniques like Grad-CAM.	Kaggle	The FLEM-XAI ensemble model achieved accuracy of 99.4% for multi-class lung disease classification.	[37]
Chest	Combined	A Federated Deep Learning framework using a custom CNN architecture integrated with XAI methods such as LIME and SHAP for model interpretation.	Chest X-ray images from distributed clients	The proposed FLPneXAINet framework achieved a high accuracy of 98.7% in detecting pneumonia from chest X-rays	[38]

Chest	Combined	A Secure Framework using MapReduce, Blockchain-based Federated Learning, and XAI with a DenseNet-201.	Multi-source CT scan datasets	The secure federated learning model achieved an exceptional accuracy of 99.8% for lung cancer prediction, leveraging blockchain for security and XAI for providing transparent, interpretable results.	[39]

2.6 Datasets Utilized in the Project

Table 2.2: Summary of Datasets Used in the Project

Dataset Name	**Domain**	**Class Distribution / Description**	**Total Images**
FracAtlas	Bone	Fractured (1707), Hand (3487), Hardware (210), Hip (796), Leg (5293), Shoulder (796)	9,383
COVID-19 & Pneumonia & Tuberculosis with Normal & Non-X-ray	Chest	COVID-19 (3,616), Normal (5,688), Pneumonia (4,861), Tuberculosis (3,453)	17,618
Chest X-ray Dataset for Lung Segmentation	Chest	Annotated chest X-ray images for lung segmentation tasks	6,106 images with corresponding 6,106 masks
Random Images for Image Classification	General	Unlabeled random images used for general image classification experiments	9,958
MURA	Bone	Elbow (1,912), Finger (2,110), Hand (2,185), Humerus (727), Forearm (1,010), Shoulder (3,015), Wrist (3,697)	40,561 images (14,656 studies)

2.7 Similar Commercial Applications

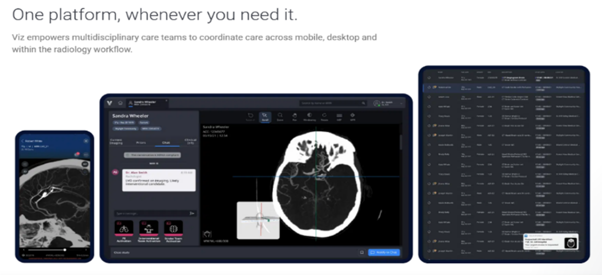
Several commercial applications leverage AI for medical image diagnosis, providing tools for clinicians to detect conditions like bone fractures and chest diseases. These applications often use DL models for classification but may lack advanced XAI or FL features for interpretability and privacy. Below, we discuss some notable examples.

2.7.1 Aidoc (aiOS™)

Aidoc is a leading clinical AI software company that utilizes its proprietary aiOS™ platform to deliver high-performance AI solutions for medical imaging interpretation and workflow optimization. The platform automatically analyzes medical scans, prioritizing critical findings like stroke, pulmonary embolism, and intracranial hemorrhage, and then notifies care teams in real-time. By integrating seamlessly with existing systems like EHR and PACS, Aidoc's solutions help radiologists streamline workflows, enhance diagnostic accuracy, and accelerate time-to-treatment for patients with acute conditions across radiology, cardiology, and neurovascular specialties.

2.7.2 Viz.ai (Viz.ai One®)

Viz.ai is a leading AI-powered care coordination platform focused on transforming time-sensitive medical care by accelerating diagnosis and treatment decisions. The Viz.ai One® platform features multiple FDA-cleared AI algorithms that analyze imaging data, including CT scans and EKGs, to detect suspected diseases like stroke and pulmonary embolism. Crucially, it then uses real-time alerts and communication tools to instantly connect multidisciplinary care teams on mobile and desktop devices, ensuring the patient receives the necessary intervention as quickly as possible.



2.7.3 Qure.ai (qXR, qCT)

Qure.ai specializes in developing deep learning algorithms for affordable and accessible diagnostics, with a strong presence in global health initiatives. Their core products, qXR (for chest X-rays) and qER/qCT (for head and lung CTs), use AI to detect and quantify abnormalities, including tuberculosis, lung nodules, and critical trauma findings. By analyzing scans with high accuracy in minutes, Qure.ai's solutions assist radiologists in early disease detection, risk stratification, and provide essential diagnostic support, especially in areas with limited access to specialists.

2.7.4 Harrison.ai

Harrison.ai, through its partnership with the I-MED Radiology Network (Annalise.ai), offers comprehensive, decision-supported AI tools designed to raise the standard of patient care and combat capacity limits. Their advanced AI solutions for Chest X-ray (CXR) and Non-contrast Head CT (CTB) are designed to detect a massive number of potential findings (often over 100) within seconds. By acting as a 'co-pilot' for clinicians, their AI solutions flag emergent and incidental findings, helping to reduce stress from chaotic worklists and significantly improving both diagnostic accuracy and efficiency.

2.7.5 Lunit (Lunit INSIGHT, Lunit SCOPE)

Lunit is a global leader in medical AI software with a singular mission to conquer cancer through cutting-edge technology. The company's solutions, including the Lunit INSIGHT suite for cancer screening (especially mammography and chest X-ray) and the Lunit SCOPE platform for precision oncology, transform complex clinical data into clear, actionable insights. By delivering highly accurate diagnostic results and extracting intelligence from tissue images to predict treatment response, Lunit empowers earlier detection and more precise treatment decisions across the entire cancer care continuum.

2.8 Summary

In this chapter, we presented the background on bone fractures and chest diseases as initial focus areas for medical image diagnosis, emphasizing multiclassification. We discussed similar commercial applications, reviewed related works on detection using DL for each domain, explainable AI techniques like GradCAM for interpretability (with potential image insertions for heatmaps), federated learning for privacy-preserving training, and combined approaches integrating these elements. We also evaluated tools for development. These insights form the foundation for our extensible project's development, highlighting gaps in interpretability and data privacy that our system aims to address for bone fracture detection, chest disease detection, and future expansions.

CHAPTER 3
PROPOSED MODELS

Chapter 3 Proposed Models

This chapter presents the proposed end-to-end solution developed for the automated detection of chest diseases and bone fractures from medical imaging data. The chapter aims to provide a comprehensive description of the overall system architecture as well as a detailed explanation of the individual models that constitute the proposed solution.

The chapter begins by outlining the solution architecture, which describes the complete pipelines of the system, including the high-level data flow and processing in the DL pipeline, and the backend processes and structure. This section illustrates how both the DL pipeline and the system's backend operate cohesively to form a unified diagnostic framework.

Subsequently, the chapter discusses the transfer learning models used in the proposed system. This section explains the pre-trained architectures employed, and their advantages in medical image analysis.

The medical image routing model is then introduced, which serves as an intelligent intermediary responsible for directing input images to the appropriate diagnostic model based on anatomical or modality-specific characteristics. This component plays a critical role in enabling the downstream models to act on an expected input image modality and anatomical region, it is also crucial in dropping out-of-distribution input data from being processed by the diagnostic models.

Following this, the bone fracture detection model is presented in detail. This section describes the model architecture, and classification approach used to identify bone fractures from radiographic images.

Then, an in-depth explanation of the chest disease detection model, which focuses on identifying various thoracic conditions from chest X-ray images, is presented. The section covers the model design, feature extraction process, and classification methodology employed to achieve robust disease detection.

Finally, the chapter ends with a summary to conclude all the information presented and an emphasis is put on the key information given in the chapter.

3.1 Solution Architecture

This section presents the architecture of the proposed system for automated detection of chest diseases and bone fractures. It describes the structural design of the solution, highlighting how its major components interact to enable end-to-end medical image analysis. The section provides a high-level overview of both the deep learning diagnostic pipeline and the supporting backend infrastructure.

3.1.1 Deep Learning Inference Pipeline

The deep learning pipeline represents the core artificial intelligence component of the proposed system and is responsible for the automated analysis of medical images. It defines the sequential flow of data from image acquisition to final diagnostic output. Figure 3.1 illustrates the overall pipeline, highlighting the interaction between the medical image routing model, and task-specific diagnostic models within a unified framework.

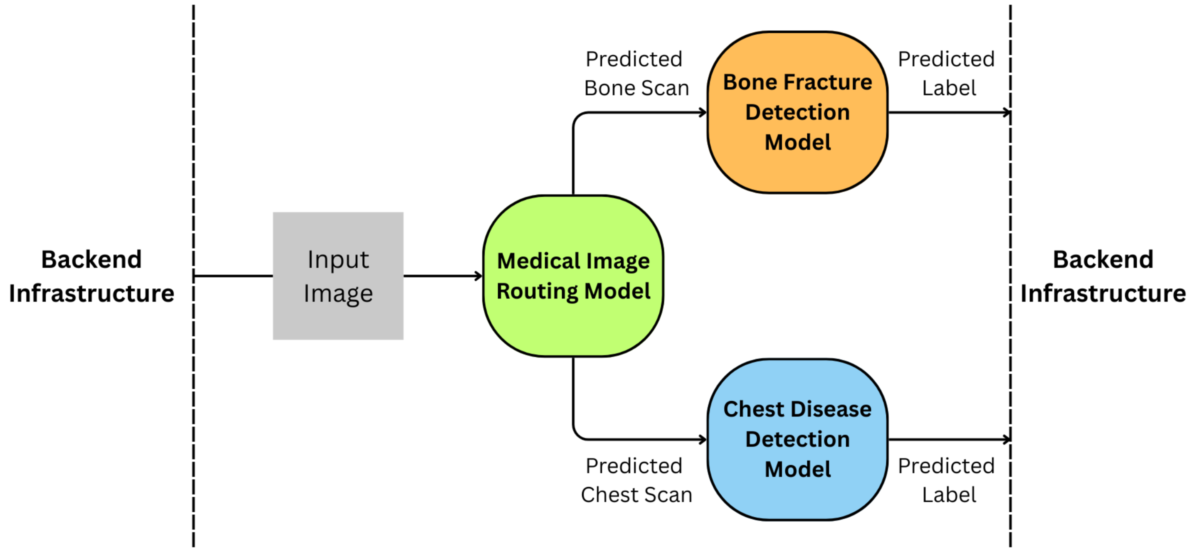


Figure 3.1: High-level diagram of the DL inference pipeline.

The pipeline accepts medical X-ray images as input. This input is coming from a user but it is firstly processed by the backend infrastructure of the solution before it is forwarded to the deep learning pipeline, this forms the basis of the assumption that all input to the pipeline is in the form of images.

The input is passed to a medical image routing model, which functions as an intelligent decision unit within the pipeline. This model identifies the anatomical category of the input image and determines the appropriate diagnostic path. By dynamically routing images to specialized models, the system achieves modularity and supports multiple diagnostic tasks within a single architecture. This unit also seizes out-of-distribution images before they are directed to the downstream diagnostic models. Out-of-distribution images represent images with an incorrect modality, from a wrong anatomical region, or non-medical images at all.

Once routed, the image is processed by a task-specific diagnostic model tailored to its medical domain. For skeletal images, the bone fracture detection model is employed, while chest X-ray images are analyzed using the chest disease detection model. Each model is independently optimized to capture domain-specific patterns relevant to its respective diagnostic objective.

The outputs of the diagnostic models are expressed as classification probabilities or predicted labels. These outputs are structured to ensure interpretability and seamless integration with downstream system components, such as backend services and user interfaces.

3.1.2 Backend Infrastructure and System Integration

The image analysis pipeline integrates user authentication, secure image upload, backend processing, and machine learning inference to deliver accurate diagnostic results. It combines a React frontend, Spring Boot backend, AWS S3 storage, and a FastAPI-based ML service to ensure efficient, scalable, and secure handling of medical images. Each step—from user login and image submission to model prediction and result visualization—is orchestrated to minimize server load, maintain data integrity, and provide interpretable outputs through Grad-CAM and YOLO visualizations, while storing all relevant metadata for future reference.

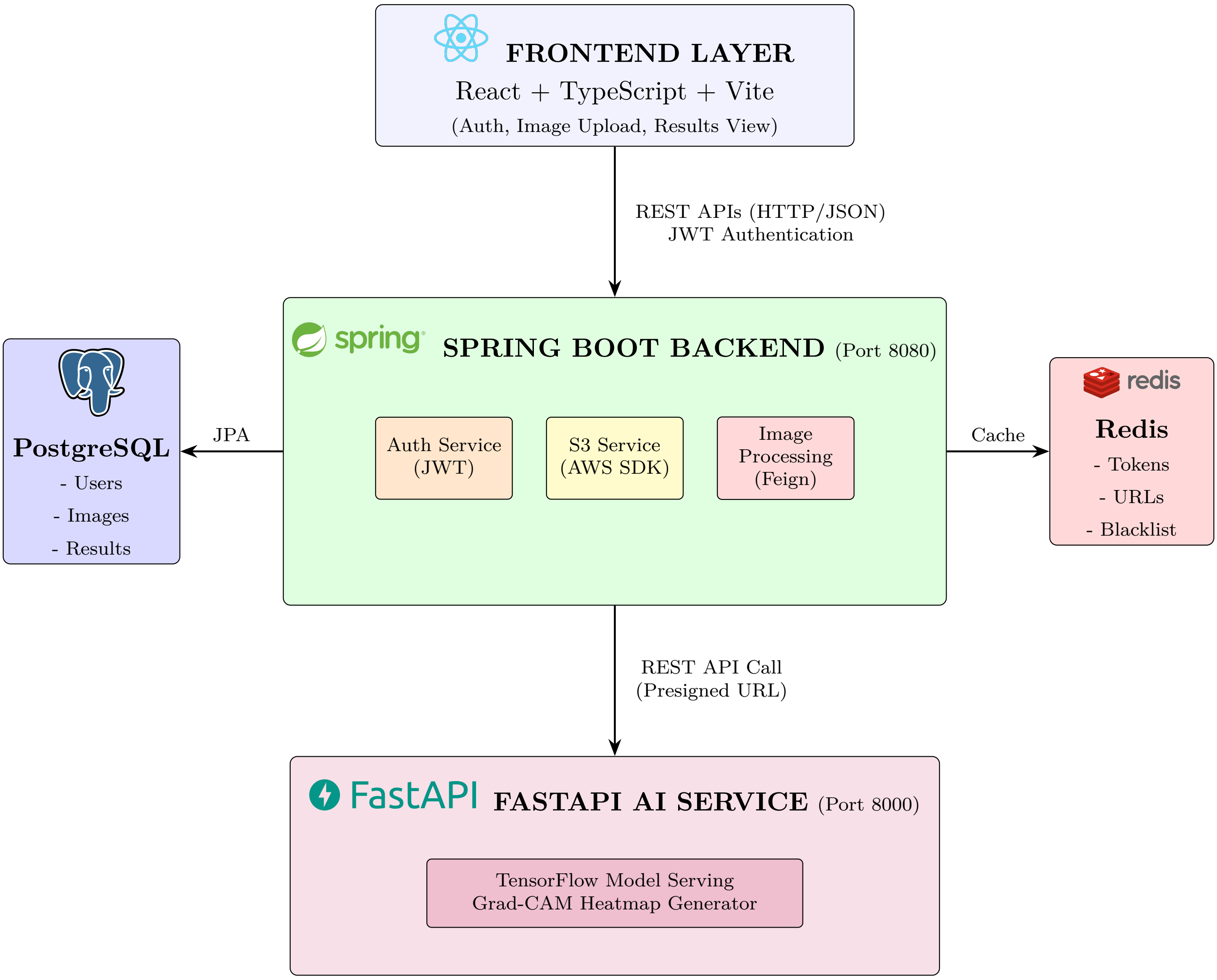


Figure 3.2: System Architecture Overview

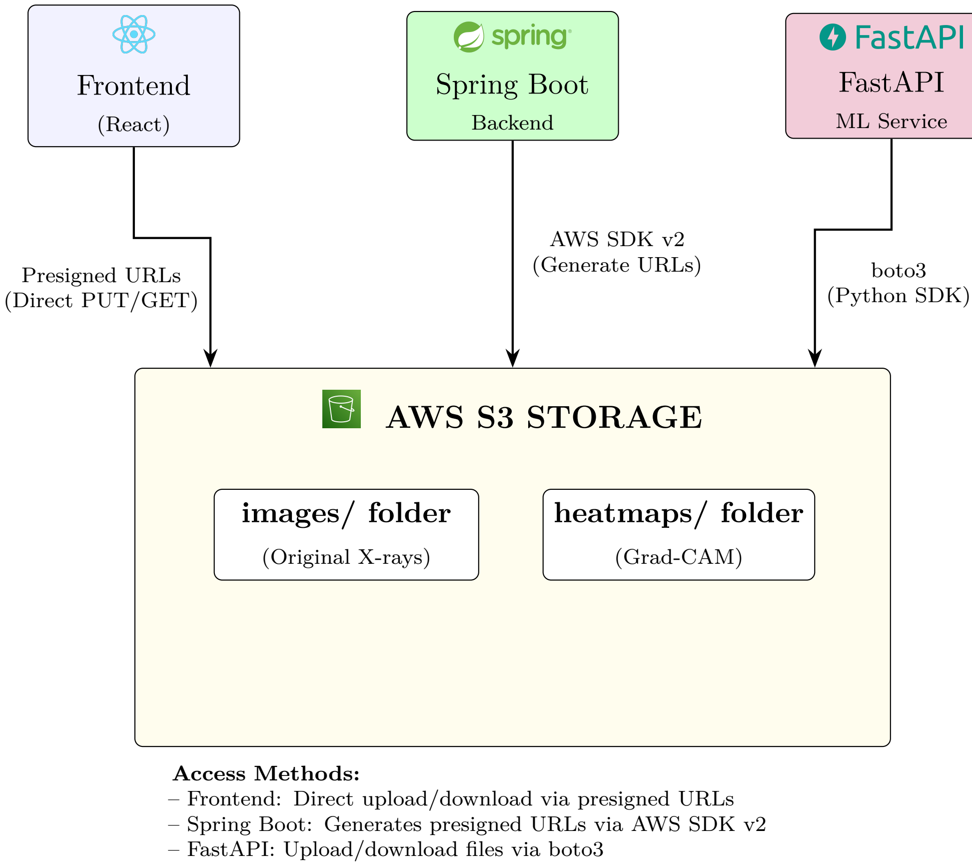


Figure 3.3: AWS S3 Integration in our system

The full system architecture diagram is shown if Figure 3.2. While the Amazon Web Services (AWS) S3 integration in the system is illustrated in Figure 3.3.

Step 1: User Authentication

The user begins by submitting their credentials through the React frontend. The frontend sends a POST request to `/auth/login`, which the Spring Boot backend validates against the PostgreSQL database. Upon successful authentication, an access token (JWT) is generated and returned in the response body, while a refresh token is stored in Redis and set as an HttpOnly cookie. The frontend stores the access token in localStorage to authenticate subsequent requests.

Step 2: Image Upload (Direct-to-S3)

For efficient image uploads, the application uses a presigned URL pattern. The frontend first requests a presigned URL by calling `POST /api/upload/presigned-url` with the filename and content type. The backend generates a unique S3 key using a UUID and creates a time-limited (5 minutes) presigned PUT URL using the AWS SDK. The frontend then uploads the image directly to S3, bypassing the backend and reducing server load.

Step 3: Image Processing Request

After uploading, the frontend initiates the processing request by calling `POST /api/image-processing/upload-and-process` with the S3 key. The backend validates the user's JWT, saves image metadata to the PostgreSQL database, and generates a presigned GET URL for the original image.

Step 4: ML Service Communication

Using Spring Cloud OpenFeign, the backend communicates with the ML service by sending a `POST /api/v1/process-image` request with the presigned URL. The ML service downloads the image from S3 and stores temporary files in a dedicated temp directory for processing.

Step 5: Model Inference

The ML service performs several operations for inference. First, the image is preprocessed: it is loaded, resized to 224×224 pixels, normalized to the $[0, 1]$ range, and its dimensions expanded for batch processing. The image is then passed through a CNN routing model, which outputs a classification of the image being in either `Bone_Scan`, `Chest_Scan`, or `Other`. After this classification is completed the image is forwarded to another model if it either a `Bone_Scan` or a `Chest_Scan`. If it is classified as `Other`, the user is informed that the input image is incorrect. The destination model is decided by the classification of the image.

For interpretability, a Grad-CAM heatmap is generated for the chest scans images, while for the bone scan images the fracture region is identified with a geometrical shape in the output image, most usually a rectangle.

Step 6: Result Storage and Response

Once inference is complete, the ML service uploads the generated image, for example a heatmap, to S3 and cleans up temporary files via a background task. The response, including the S3 key, prediction label, and confidence score, is sent back to Spring Boot. The backend saves the result in the database, generates a presigned URL for the heatmap, and returns the complete response to the frontend.

Step 7: Result Display

On the frontend, the prediction results and presigned URLs are used to display the original X-ray and the generated image side-by-side. The predicted label and confidence percentage are shown to the user, and the results are stored in the prediction history for future reference.

3.2 Transfer-Learning Models Used

This section presents the transfer learning models employed in the proposed system and discusses the rationale behind their selection. Transfer learning is adopted to leverage pre-trained deep neural networks that have been trained on large-scale datasets, enabling effective feature extraction and improved generalization in medical imaging tasks with limited labeled data. The section provides an overview of the selected architectures and highlights their suitability for different components of the proposed diagnostic pipeline.

3.2.1 ResNet50 Model

Residual Networks (ResNets) were introduced to address the degradation problem observed in deep convolutional neural networks, where increasing network depth leads to saturation or decline in training accuracy. This issue emerges due to optimization difficulties such as vanishing gradients. ResNet addresses this limitation through the concept of residual learning, which allows the training of substantially deeper networks by enabling more effective gradient propagation.

The key idea of residual learning is to reformulate the learning objective of a neural network block. Instead of directly learning a desired underlying mapping $H(x)$, a residual block learns a residual function $F(x)$ such that:

$$H(x) = F(x) + x$$

where x represents the input to the residual block and $F(x)$ represents the output of the stacked convolutional layers within the block. The identity shortcut connection that performs the addition allows gradients to flow directly through the network during backpropagation, thereby minimizing vanishing gradient effects and improving training stability.

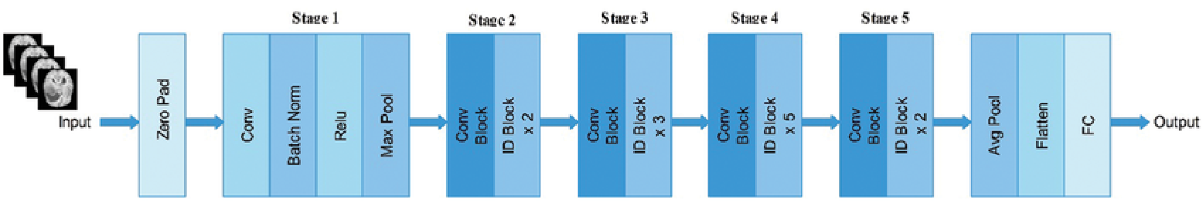


Figure 3.4: Architecture of the ResNet50 illustrating the successive stages of the model.

As illustrated in Figure 3.4, the ResNet50 architecture begins with an initial convolutional layer preceded by zero padding, followed by batch normalization, a ReLU activation function, and a max pooling operation. This initial stage is responsible for extracting low-level visual features from the input image. The network then proceeds through five sequential stages composed of convolutional blocks and identity blocks, each incorporating residual connections that bypass multiple convolutional layers.

ResNet50 employs a bottleneck design within its residual blocks, consisting of a sequence of 1×1, 3×3, and 1×1 convolutional layers. The 1×1 convolutions are used to reduce and subsequently restore feature dimensionality, thereby decreasing computational complexity while preserving representational power. This design enables ResNet50 to achieve substantial depth without incurring prohibitive computational costs.

Compared to shallower variants such as ResNet18 and ResNet34, ResNet50 offers increased representational capacity and improved ability to capture complex and abstract visual patterns. While deeper variants such as ResNet101 and ResNet152 further extend network depth, they introduce higher computational overhead and an increased risk of overfitting, particularly in medical imaging applications where annotated data is often limited. Consequently, ResNet50 represents a balanced trade-off between depth, efficiency, and performance within the ResNet family.

As shown in Figure 3.4, the final stages of ResNet50 consist of a global average pooling layer, followed by feature flattening and a fully connected layer. In transfer learning scenarios, this classification head can be replaced or fine-tuned to adapt the network to domain-specific medical tasks.

In the context of medical image analysis, ResNet50 is well-suited for extracting both low-level structural features and high-level semantic representations relevant to pathological patterns. The residual connections promote feature reuse and robust optimization, which is particularly important for detecting subtle abnormalities in medical images such as fractures or thoracic diseases. These characteristics make ResNet50 a reliable and widely adopted backbone for medical image classification tasks.

3.2.2 YOLOv8 Model

As shown in Figure 3.5, the architecture is a "single-stage" pipeline that processes an X-ray in one forward pass, making it fast eno ugh for real-time clinical use. It is composed of three main parts:

- **The Backbone (C2f Module):** This is the model's feature extractor. It uses the C2f (Cross-stage Partial Bottleneck with 2 convolutions) structure to scan the X-ray for textures. It learns to distinguish the high-density, sharp-edged appearance of Hardware (screws/plates) from the organic textures of Hand, Hip, Leg, or Shoulder bones.
- **The Neck (PANet):** This part handles "feature fusion." In medical imaging, context is everything. The Neck combines fine details (like a jagged fracture line) with global shapes (the overall structure of a Hip vs. a Shoulder). This helps the model realize that a specific shadow is a Fractured region and not just a natural bone groove.
- **The Head (Decoupled & Anchor-Free):** Unlike older models, YOLOv8 uses a Decoupled Head, which separates the task of identifying the class (e.g., "Leg") from the task of drawing the box. It is Anchor-Free, meaning it predicts the center of the object directly. This is much better for medical use because fractures and surgical hardware don't come in "standard" box shapes.

During training, the model uses three mathematical "grades" to improve its accuracy:

- **Classification Loss (BCE):** Using Binary Cross-Entropy, the model learns to ac-

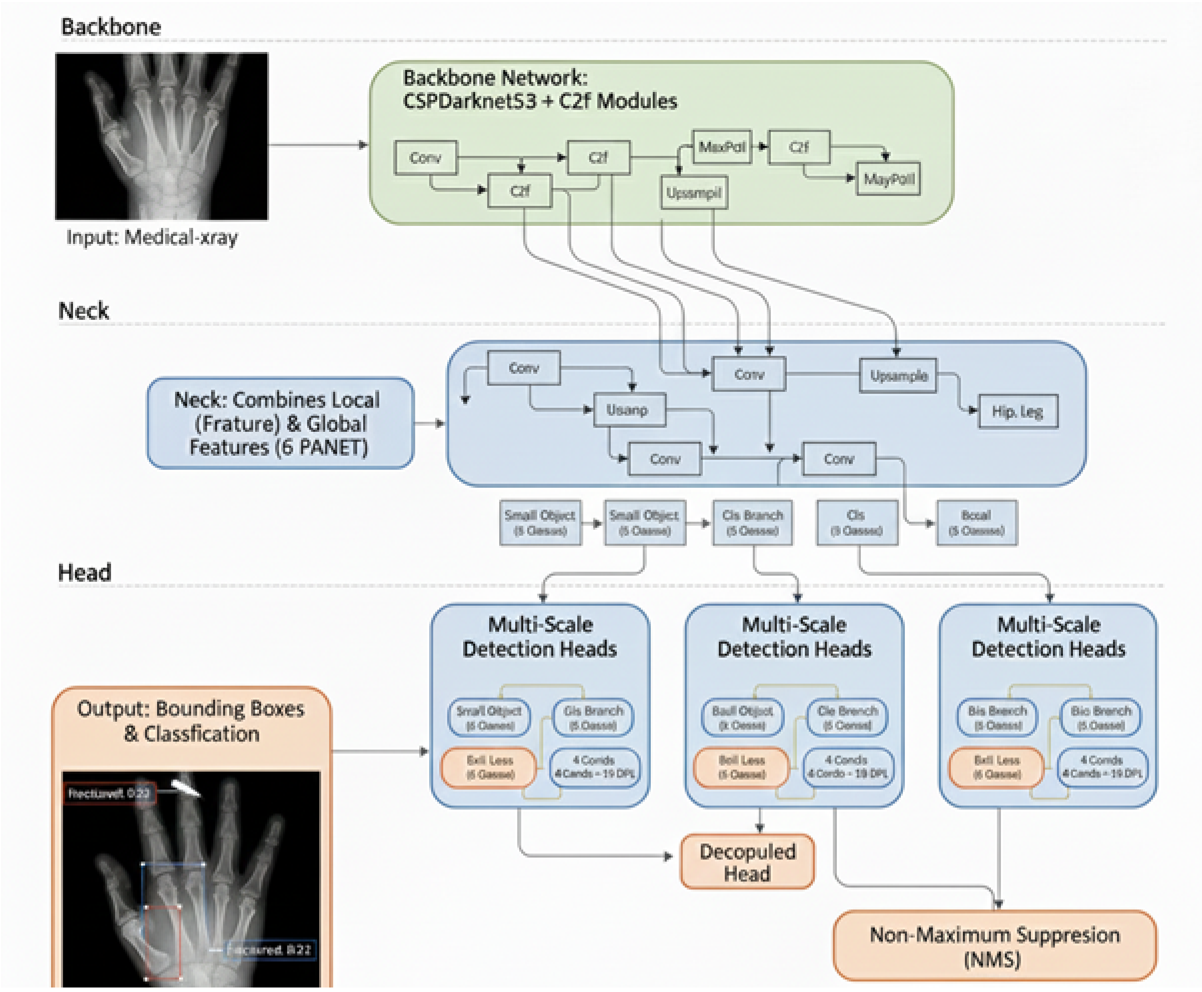


Figure 3.5: Architecture of the YOLOv8 illustrating the successive stages of the model.

curately label its 6 classes. It ensures that 'Hardware' isn't confused with 'Fractured' bone and that the anatomical regions (Hand, Hip, etc.) are correctly identified.

- **Box Localization Loss (CIoU):** This measures how well the predicted box overlaps with the actual object. For a long Leg bone or a small Hardware screw, CIoU adjusts the box's aspect ratio and center point to ensure a tight, perfect fit.
- **Distribution Focal Loss (DFL):** This is critical for hairline fractures. Since the edges of a crack can be blurry on an X-ray, DFL predicts a probability distribution for the box boundaries, allowing the model to be more precise even when the image quality is low.

To ensure the model is both reliable and clinically safe, we evaluate its performance using three primary metrics: Precision, Recall, and mAP50. These indicators provide a comprehensive look at the model's accuracy, its ability to detect every injury, and its overall effectiveness in a real-world medical environment. By analyzing these figures, we can understand how the system balances the need for "Quality Control" with the "Safety Net" required to ensure no fracture goes unnoticed.

Precision acts as the model's primary "Quality Control" mechanism. In the context of medical imaging, high precision ensures that when the model identifies an injury as

"Fractured," there is a high statistical certainty that a break actually exists. By focusing on the accuracy of positive predictions, precision prevents doctors from being overwhelmed by "False Alarms," allowing them to trust that the alerts they receive are legitimate and actionable.

While precision focuses on accuracy, Recall serves as the critical "Safety Net" for patient care. It measures the model's ability to find every single instance of a fracture within the dataset. Maintaining high recall is vital in a clinical setting because it ensures that no patient is accidentally sent home with an undiagnosed broken bone. In short, recall prioritizes "catching" every case, minimizing the risk of a dangerous false negative.

The mAP50, or mean Average Precision at a 50% overlap threshold, serves as the comprehensive "Overall Score" for the model's performance. This metric provides the best single indicator of how well the model identifies both the specific anatomy and the underlying injuries simultaneously. By calculating the average precision across all categories at a standard detection threshold, mAP50 offers a balanced view of the system's total effectiveness.

3.3 Medical Image Routing Model

This section presents the proposed medical image routing model employed in the system and discusses the motivation behind the model, its problem formulation, and finally its architecture. This model serves as the decision-making unit of the system, where the destination of each input image is determined and forwarded thereafter.

3.3.1 Motivation

The increasing use of deep learning in medical image analysis has enabled the development of highly specialized models for tasks such as disease detection and diagnosis. In the context of lung cancer detection, chest X-ray and computed tomography (CT) images provide complementary diagnostic information and often require different processing pipelines and model architectures. While chest X-rays offer a low-cost and widely accessible imaging modality for preliminary screening, CT scans provide detailed cross-sectional information that supports more comprehensive assessment of lung abnormalities.

However, in real-world deployment scenarios, diagnostic systems are often exposed to heterogeneous image inputs originating from different imaging modalities, anatomical regions, or acquisition protocols. Applying a modality-specific diagnostic model to an inappropriate image type may lead to unreliable predictions and compromise the robustness and safety of the overall system. Consequently, there is a need for an initial classification mechanism capable of identifying the nature of incoming medical images before further analysis is performed.

This section introduces a medical image classification model designed to function as an image routing component within a larger lung cancer diagnostic pipeline. The model distinguishes between chest X-ray images, chest CT scan images, and an additional out-of-distribution category referred to as “Other.” By incorporating this third category, the system is able to explicitly handle images that do not belong to the intended diagnostic classes, thereby reducing the risk of misclassification and improving overall system reliability. Once classified, chest X-ray and CT scan images are forwarded to their corresponding lung cancer diagnostic models, while out-of-distribution images are rejected from further analysis. The following subsections present the motivation for this approach, formally define the classification problem, and describe the model architecture employed.

3.3.2 Problem Statement

The goal of this work is to develop an automated medical image classification model capable of identifying the type of an input image and routing it to the appropriate downstream analysis. The task is formulated as a three-class classification problem, where each input image $X \in \mathbb{R}^{224\times224\times3}$ must be assigned to one of three categories: chest X-ray images (0), chest CT scan images (1), or out-of-distribution images labeled as *Other* (2). This multi-class formulation ensures that irrelevant or ambiguous inputs are explicitly handled, improving system safety and reducing the risk of inappropriate downstream processing.

The model is designed to output a probability distribution $\mathbf{p} = [p_0, p_1, p_2]$ over the three classes, where $p_k = P(y = k \mid X)$ and $\sum_{k=0}^{2} p_k = 1$. The predicted class is obtained by selecting the class with the highest probability, $\hat{y} = \arg\max_k p_k$. Images classified as chest X-rays are forwarded to the X-ray-based lung cancer detection model, while images classified as CT scans are routed to the CT-based lung cancer detection model. By incorporating the *Other* category, the model avoids forcing all inputs into diagnostic categories and can robustly separate images outside the target distribution, laying a solid foundation for safe integration within a larger medical image analysis pipeline.

3.3.3 Model Architecture

The proposed medical image classification model is designed using a transfer learning approach based on the ResNet50 convolutional neural network. The architecture follows a structured pipeline consisting of input preprocessing, deep feature extraction, feature aggregation, and final classification. This modular design allows the model to benefit from representations learned on large-scale image datasets while adapting effectively to the target medical imaging task. An overview of the complete classification pipeline is illustrated in Figure 3.6.

The input to the model is a medical image $X \in \mathbb{R}^{224\times224\times3}$, resized to a fixed spatial resolution and represented in RGB color space. Prior to being processed by the pretrained network, the image undergoes a preprocessing step to ensure compatibility with the ResNet50 weights. This step mirrors the preprocessing applied during the original ImageNet training of ResNet50 and is essential for effective transfer learning. Specifically, the input image is first converted to floating-point representation, its color channels are reordered from RGB to BGR, and a fixed channel-wise mean is subtracted. Let X_{rgb} denote the raw input image; the preprocessed image X_{prep} is obtained as $X_{prep} = \mathrm{BGR}(X_{rgb}) - \mu$, where $\mu = [103.939, 116.779, 123.68]$ represents the ImageNet mean for the blue, green, and red channels, respectively. This operation centers the input distribution around zero and aligns it with the statistical properties expected by the pretrained backbone.

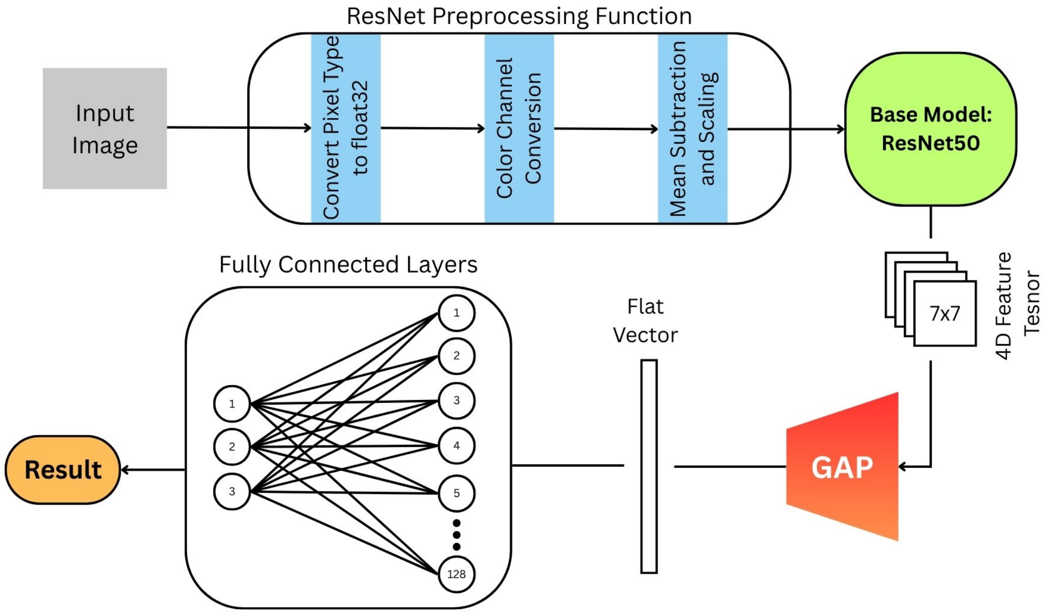


Figure 3.6: Overall architecture of the proposed medical image routing model.

Following preprocessing, the image is passed through the ResNet50 network with its original classification layers removed. In this configuration, ResNet50 functions as a fully convolutional feature extractor. For an input tensor of shape $(224, 224, 3)$, the output of the backbone is a high-dimensional feature map of shape $(7, 7, 2048)$. This tensor contains 2048 feature channels, each encoding abstract visual patterns such as anatomical structures, textures, and modality-specific characteristics while preserving coarse spatial relationships within the image. For a detailed discussion of the ResNet50 model architecture, refer to Subsection 3.2.1.

To transform the spatial feature maps into a compact representation suitable for classification, a global average pooling operation is applied to the output of the ResNet50 backbone. This operation computes the average activation of each feature channel across the spatial dimensions, effectively summarizing the presence of each learned feature over the entire image. Given a feature map $F \in \mathbb{R}^{7\times7\times2048}$, the pooled feature vector $z \in \mathbb{R}^{2048}$ is computed such that each element z_k corresponds to the mean of the activations in channel k, i.e.

$$z_k = \frac{1}{7 \times 7} \sum_{i=1}^{7} \sum_{j=1}^{7} F_{i,j,k}$$

This operation reduces the tensor shape from $(7, 7, 2048)$ to (2048) while significantly lowering the number of parameters required in subsequent layers and improving generalization.

The resulting feature vector is then passed to a task-specific classification head composed of fully connected layers. These layers learn discriminative combinations of the extracted features and map them to the target class space. The final layer applies a softmax activation function, producing a probability distribution $\mathbf{p} = [p_0, p_1, p_2]$, where each component represents the predicted probability of the image belonging to the chest X-ray class, chest CT scan class, or the out-of-distribution *Other* class. The predicted label is obtained as $\hat{y} = \arg\max_k p_k$.

3.4 Bone Fracture Detection Model

This section describes the deep learning model employed for automated bone fracture detection within the proposed system. The objective of this model is to accurately identify fracture patterns in skeletal X-ray images as part of the end-to-end diagnostic pipeline. Given the real-time requirements and spatial nature of fracture localization, a one-stage object detection framework was selected. To avoid redundancy, this section focuses on the task-specific motivation and formulation, while architectural details already discussed in Section X are referenced where appropriate.

3.4.1 Motivation

Bone fractures are among the most common musculoskeletal injuries and require timely and accurate diagnosis to prevent complications. Manual fracture detection from radiographic images is a time-consuming process that is highly dependent on the expertise of the radiologist and may be prone to inter-observer variability. Automated fracture detection systems have the potential to assist clinicians by providing fast and consistent assessments, particularly in emergency and high-throughput clinical settings.

From a technical perspective, fracture detection is inherently a localization problem, as fractures often occupy small and irregular regions within an image. Therefore, a model capable of both identifying and localizing regions of interest is essential. YOLOv8 was selected due to its ability to perform object detection in a single forward pass, offering an effective balance between detection accuracy and computational efficiency.

3.4.2 Problem Statement

The bone fracture detection task is formulated as a supervised object detection problem. Given an input skeletal X-ray image, the objective is to determine the presence and location of bone fractures by predicting bounding boxes and corresponding class labels. The model is trained to distinguish fractured regions from normal bone structures, accounting for variations in fracture type, size, orientation, and imaging conditions.

Let an input image be denoted by I, and the model output be represented as a set of predictions:

$$\mathcal{P} = \{(b_i, c_i, s_i)\}_{i=1}^{N}$$

where b_i denotes the predicted bounding box, c_i represents the fracture class label, and s_i is the confidence score associated with the prediction. The goal of the model is to maximize detection accuracy while minimizing false positives, particularly in clinically sensitive regions.

3.4.3 Model Architecture

The bone fracture detection model is implemented using the YOLOv8 architecture without any structural modifications or additional components. The pre-trained YOLOv8 model is adopted directly and fine-tuned on the fracture dataset to adapt its learned representations to the skeletal imaging domain. No custom layers, architectural changes, or task-specific alterations were introduced.

YOLOv8 follows a one-stage detection paradigm consisting of a backbone network for feature extraction, a neck module for multi-scale feature aggregation, and a detection head responsible for predicting bounding boxes, class probabilities, and confidence scores. This architecture enables efficient detection of fractures at multiple scales, which is particularly important given the varying size and appearance of fracture patterns in X-ray images.

Since YOLOv8 has already been described in detail in Subsection 3.2.2, a comprehensive architectural discussion is not repeated here. Instead, the reader is referred to the previously presented description and to the official YOLOv8 documentation and reference literature for an in-depth explanation of the model design, training strategy, and loss functions. The use of the unmodified YOLOv8 architecture ensures reproducibility, robustness, and alignment with established object detection benchmarks.

Several attempts have been made to utilize the DenseNet121 model is this classification task. The first trial was done on the MURA dataset whose properties are briefly described in Table2.2, it yielded an accuracy of 79.9% which was a considerable improvement over the results yielded by the ResNet50 model when used on the same dataset, ResNet50 yielded an accuracy of only 60%, but it still did not achieve the project objective outlined in Section 1.6. A similar attempt to use the DenseNet121 model was used on the FracAtlas dataset which was also described in Table 2.2, and although it yielded acceptable quantitative metrics, after employing Grad-CAM for interpretbility, it was discovered that the model did not learn from the fracture areas at all but from features in other areas. Thus, after these trials the DenseNet121 model was discarded and we focused on utilizing the YOLOv8 model instead.

Figure 3.7 shows one of the preprocessing pipelines employed in the attempts to utilize the DenseNet121 model.

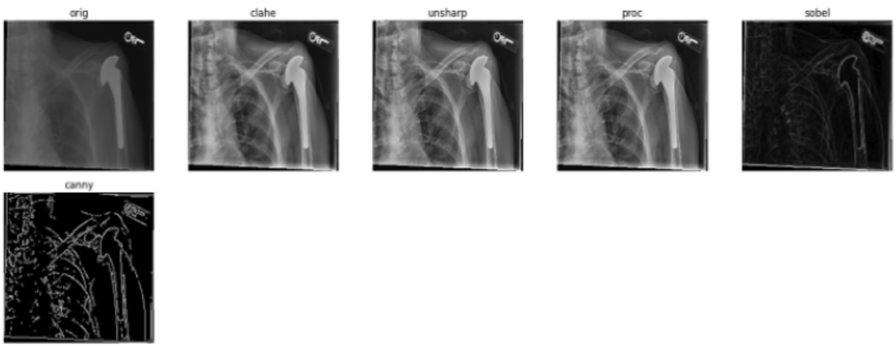


Figure 3.7: Image preprocessing pipeline used in the attempt to train DenseNet121 on the MURA dataset.

3.5 Chest Disease Classification Model

This section presents one of the downstream diagnostic models developed within the project, focusing on automated multi-class classification of chest diseases from chest X-ray (CXR) images. The model is designed as a robust and interpretable pipeline that integrates standardized image preprocessing, lung region isolation, and deep learning–based classification using transfer learning. Emphasis is placed on architectural design choices and methodological foundations rather than performance evaluation, which is discussed in a later chapter.

3.5.1 Motivation

Deep learning has demonstrated strong potential in medical image analysis, yet several challenges remain, including variability in image acquisition, class imbalance, and the need for explainable decision-making. The proposed model addresses these issues through a multi-stage pipeline that enhances image quality, isolates clinically relevant lung regions, and applies a transfer learning–based classifier augmented with explainability techniques.

3.5.2 Problem Statement

The objective of this model is to perform multi-class classification of chest X-ray images into the following diagnostic categories:

$$\mathcal{C} = \{\text{NORMAL}, \text{PNEUMONIA}, \text{COVID-19}, \text{TUBERCULOSIS}\}.$$

Given an input chest X-ray image $I \in \mathbb{R}^{H\times W}$, the task is to learn a mapping

$$f_\theta : I \to \mathbf{p},$$

where $\mathbf{p} \in \mathbb{R}^{|\mathcal{C}|}$ is a probability distribution over disease classes and θ denotes the model parameters.

To improve robustness and clinical relevance, the classification task is constrained to lung parenchyma regions extracted via an explicit segmentation stage. Additionally, the model must provide interpretable outputs to support clinical trust, motivating the use of gradient-based visualization techniques.

3.5.3 Model Architecture

The diagnostic model is structured as a multi-stage pipeline, illustrated conceptually in Figure 3.8. The pipeline consists of three main components: image preprocessing, lung segmentation, and disease classification.

Each chest X-ray image undergoes a standardized preprocessing pipeline to enhance diagnostically relevant features. Images are first converted to grayscale and enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). Noise introduced during enhancement is mitigated using Gaussian smoothing with a 5×5 kernel.

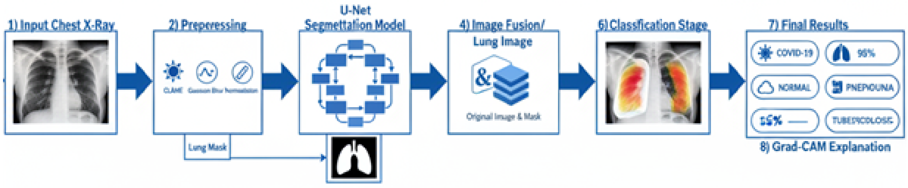


Figure 3.8: High-level overview of the multi-stage chest disease classification pipeline.

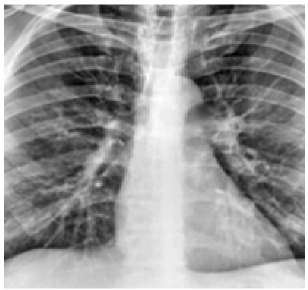
Pixel intensities are then normalized using Min–Max normalization: $I_{\text{norm}} = \frac{I - I_{\text{min}}}{I_{\text{max}} - I_{\text{min}}}$ where I_{min} and I_{max} represent the minimum and maximum pixel values of the image. Finally, images are converted to RGB format to ensure compatibility with the ResNet50 model. For a discussion of the ResNet50 model, refer to Subsection 3.2.1.

Figure 3.9 illustrates the effect of the preprocessing pipeline on representative CXR images.

(a) Original Pneumonia CXR

(b) Preprocessed Pneumonia CXR

(c) Original Pneumonia CXR



(d) Preprocessed Pneumonia CXR

Figure 3.9: Chest X-ray images before and after preprocessing, highlighting improved contrast and reduced noise.

To ensure that classification focuses exclusively on clinically relevant regions, a U-Net–based segmentation model is employed to extract lung masks. The U-Net architecture follows an encoder–decoder structure with skip connections that preserve spatial information, making it well suited for biomedical image segmentation. The U-Net architecture used is shown in Figure 3.10

The segmentation network outputs a binary lung mask $M \in \{0,1\}^{H \times W}$, which is applied to the preprocessed image using element-wise multiplication:

$$I_{\text{lung}} = I_{\text{norm}} \odot M.$$

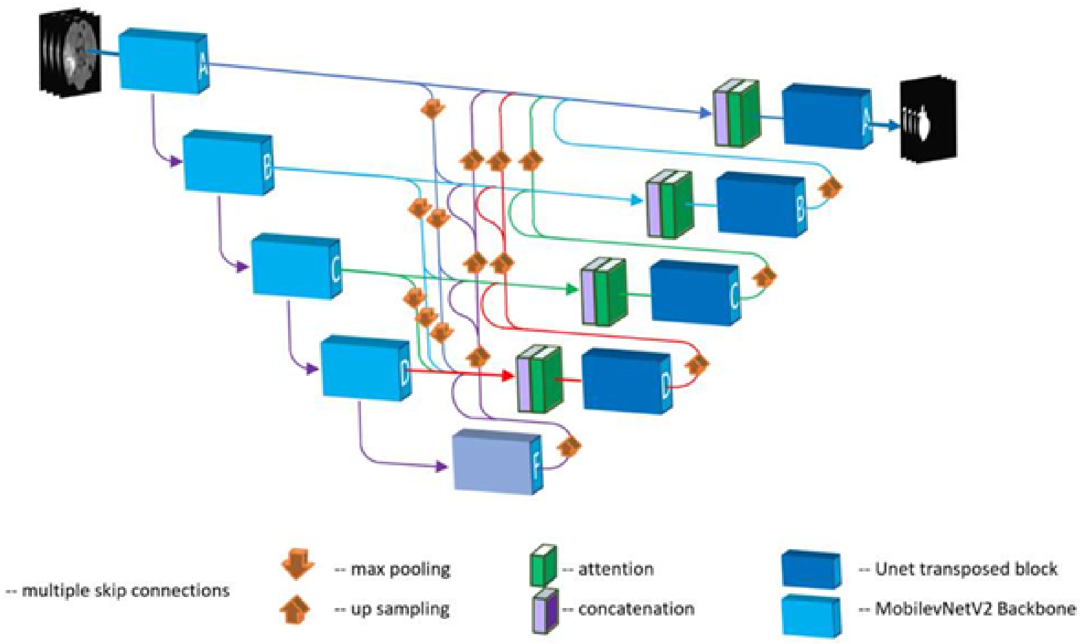


Figure 3.10: The U-Net architecture, illustrating the encoder-decoder structure with skip connections that merge high-resolution features from the contracting path with upsampled features in the expansive path. This design is highly effective for precise medical image segmentation.

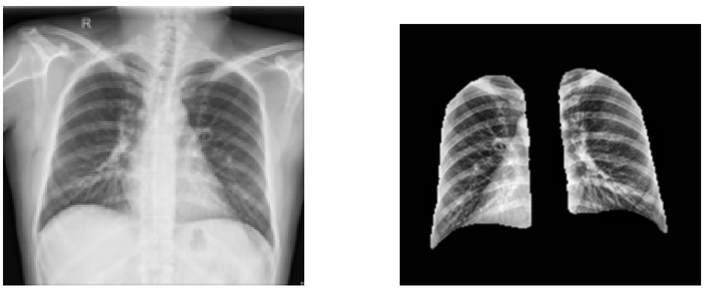
Training of the segmentation model is guided by the Dice Loss, derived from the Dice–Sørensen Coefficient (DSC):

$$\text{DSC} = \frac{2|X \cap Y|}{|X| + |Y|},$$

where X and Y denote the predicted and ground-truth masks, respectively. The loss function is defined as

$$\mathcal{L}_{\text{Dice}} = 1 - \text{DSC},$$

which is particularly effective in handling class imbalance between lung and background pixels. The result of using segmentation on CXR images is illustrated in Figure 3.11



(a) Preprocessed CXR Image

(b) Segmented Lung Image after Mask Application

Figure 3.11: Application of the U-Net segmentation mask. The original preprocessed Figure 3.11a is processed to produce the final segmented lung Figure 3.11b, which serves as the input for the classification stage

The final classification stage employs a ResNet50-based transfer learning model. ResNet architectures leverage residual connections to facilitate gradient flow and enable the training of deep networks. As the ResNet family is discussed in detail earlier in this book, only a brief overview is provided here.

A ResNet50 model pre-trained on ImageNet is used as a feature extractor, with early layers frozen to preserve general visual representations. A custom classification head is appended, consisting of global average pooling, fully connected layers with ReLU activation, dropout for regularization, and a final softmax layer producing class probabilities.

The model is trained using Sparse Categorical Cross-Entropy loss:

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{i=1}^{N} \log p_{i,c},$$

where $p_{i,c}$ denotes the predicted probability of the true class c for sample i. To address class imbalance, class-weighted loss scaling is applied during training.

To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to visualize the regions most influential to model predictions, an example is shown in Figure 3.12. Grad-CAM computes the gradients of the target class score with respect to the feature maps of the final convolutional layer and produces a coarse localization map highlighting salient regions.

These heatmaps are overlaid on the original chest X-ray images, enabling qualitative assessment of whether the model attends to clinically meaningful lung regions.

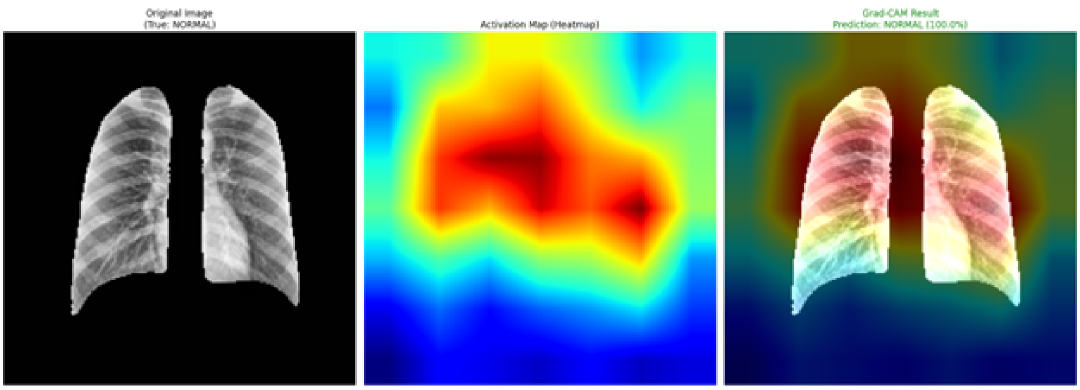


Figure 3.12: Grad-CAM visualizations highlighting regions contributing to model predictions.

3.6 Summary

This chapter presented the proposed end-to-end model for automated detection of chest diseases and bone fractures from medical images. The overall solution architecture was introduced, highlighting the integration of a deep learning–based diagnostic pipeline with supporting backend processes to enable efficient and scalable medical image analysis.

The deep learning pipeline was described in detail, illustrating the sequential flow from image preprocessing and feature extraction to intelligent routing and task-specific inference. Transfer learning was adopted as a core strategy to leverage pre-trained deep neural networks, enabling effective feature representation and improved generalization in the presence of limited labeled medical data.

The chapter also introduced the medical image routing model, which enables dynamic task allocation by directing input images to specialized diagnostic models. This design enhances modularity and allows the system to support multiple medical imaging tasks within a unified framework.

Then, the architecture of the chest disease classification model was discussed in detail. The design was illustrated, and the use of ResNet50 as the main feature extractor was justified. The formulation of the problem and the motivation behind such a model were presented while detailed architectural aspects of the transfer-learning model were omitted to avoid redundancy.

Finally, the bone fracture detection model was presented as a task-specific component of the system, implemented using an unmodified YOLOv8 architecture. The formulation of the fracture detection problem and the motivation for adopting a one-stage object detection approach were discussed, while detailed architectural aspects were referenced to earlier sections to avoid redundancy.

Overall, this chapter established the structural and methodological foundation of the proposed system, providing a clear understanding of its components and design choices. The described models form the basis for the experimental evaluation and performance analysis presented in the subsequent chapter.

CHAPTER 4 RESULTS & EVALUATION

Chapter 4 Results and Evaluation

This chapter presents preliminary results and an initial evaluation of the proposed models. The current version of the chapter is intended as a draft to demonstrate that the models have been successfully developed, trained, and integrated within the proposed system, and that a subset of the required experimental results has already been obtained. While the presented findings provide early evidence of the feasibility and effectiveness of the proposed approach, the evaluation will be expanded and refined in the final version of the manuscript to include additional experiments, metrics, and comparative analyses.

4.1 Evaluation Metrics

To quantitatively evaluate the performance of the proposed models, standard classification metrics are employed, namely precision, recall, accuracy, and F1-score. These metrics are computed based on the confusion matrix, which consists of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

Precision measures the proportion of correctly predicted positive samples among all samples predicted as positive and is defined as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall, also referred to as sensitivity, measures the proportion of correctly predicted positive samples among all actual positive samples and is given by:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Accuracy represents the overall correctness of the model by measuring the proportion of correctly classified samples among all samples and is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

The F1-score provides a single metric that balances precision and recall by computing their harmonic mean. It is particularly useful in scenarios involving class imbalance, which are common in medical imaging applications. The F1-score is defined as:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Together, these metrics offer a comprehensive evaluation of model performance. Precision reflects the model's ability to minimize false positive predictions, recall evaluates its effectiveness in identifying positive cases, accuracy provides an overall measure of correctness, and the F1-score captures the trade-off between precision and recall. This combination of metrics ensures a balanced and reliable assessment of the proposed models across different diagnostic tasks.

4.2 Results

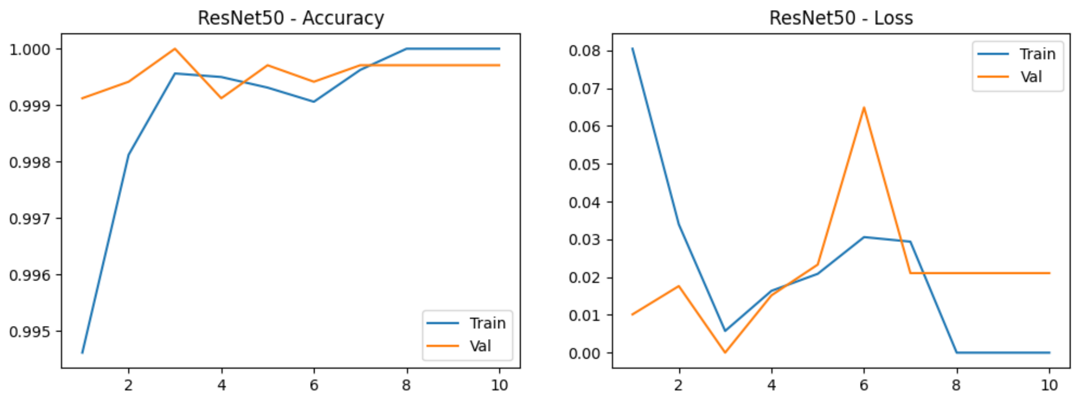


Figure 4.1: Training dynamics of the medical image routing model.

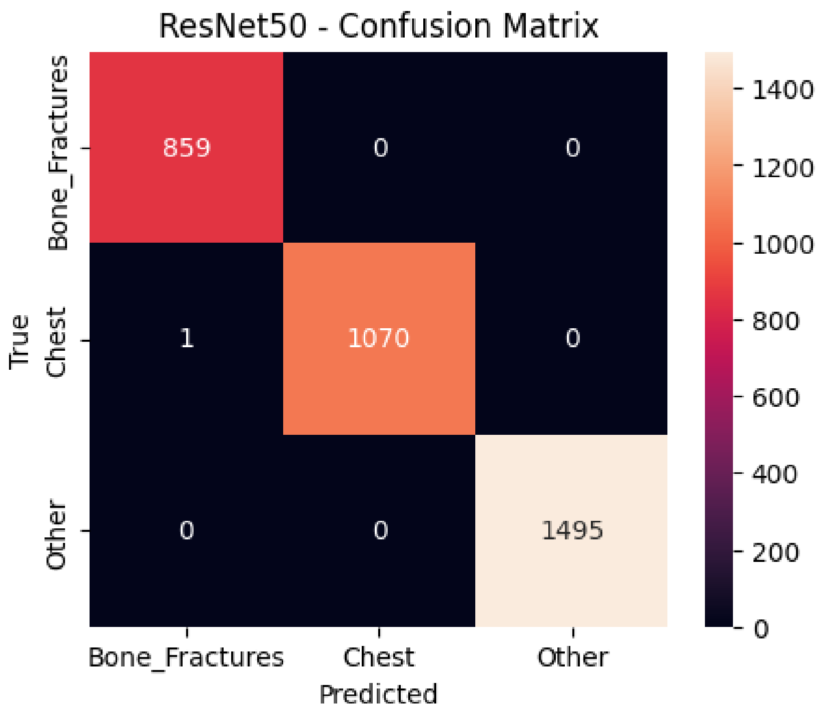


Figure 4.2: Confusion matrix of the medical image routing model.

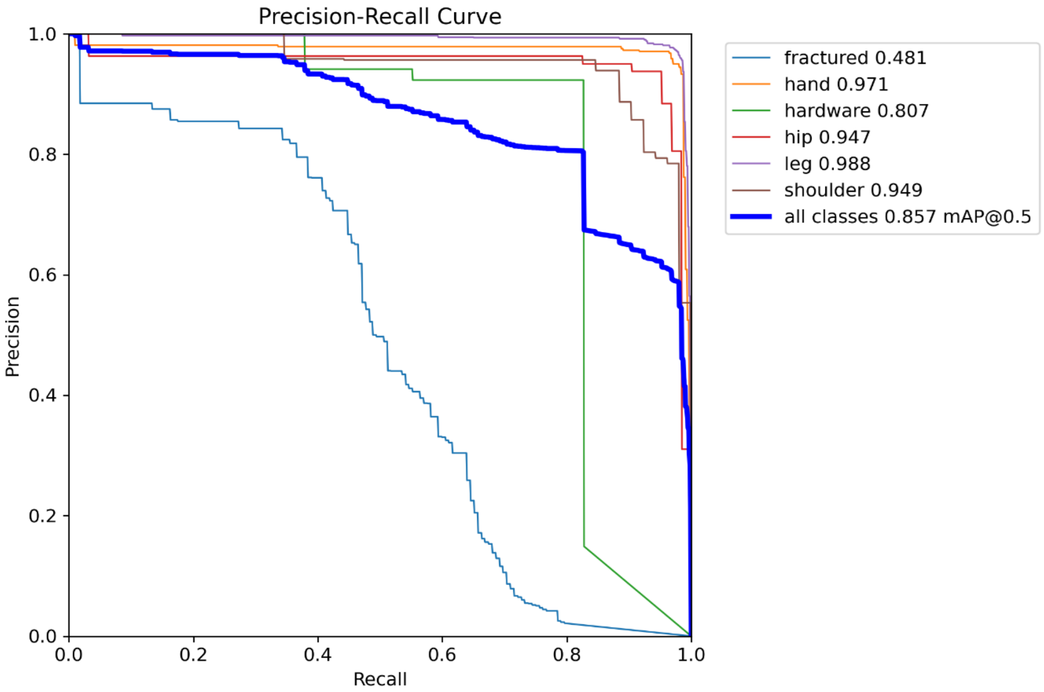


Figure 4.3: Bone fracture detection model Recall-Precision curve.

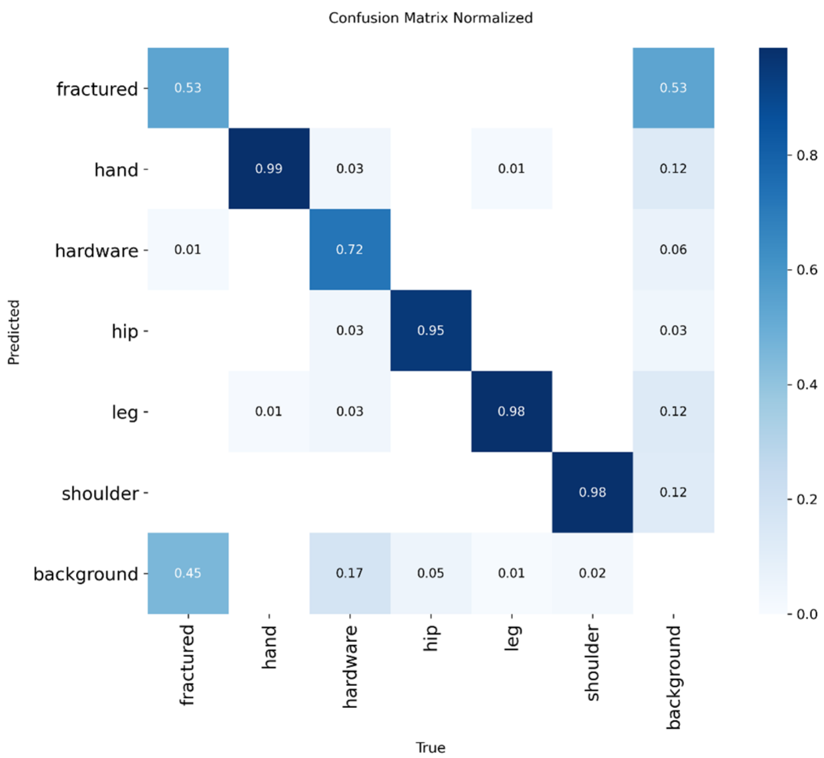


Figure 4.4: Bone fracture detection model confusion matrix.

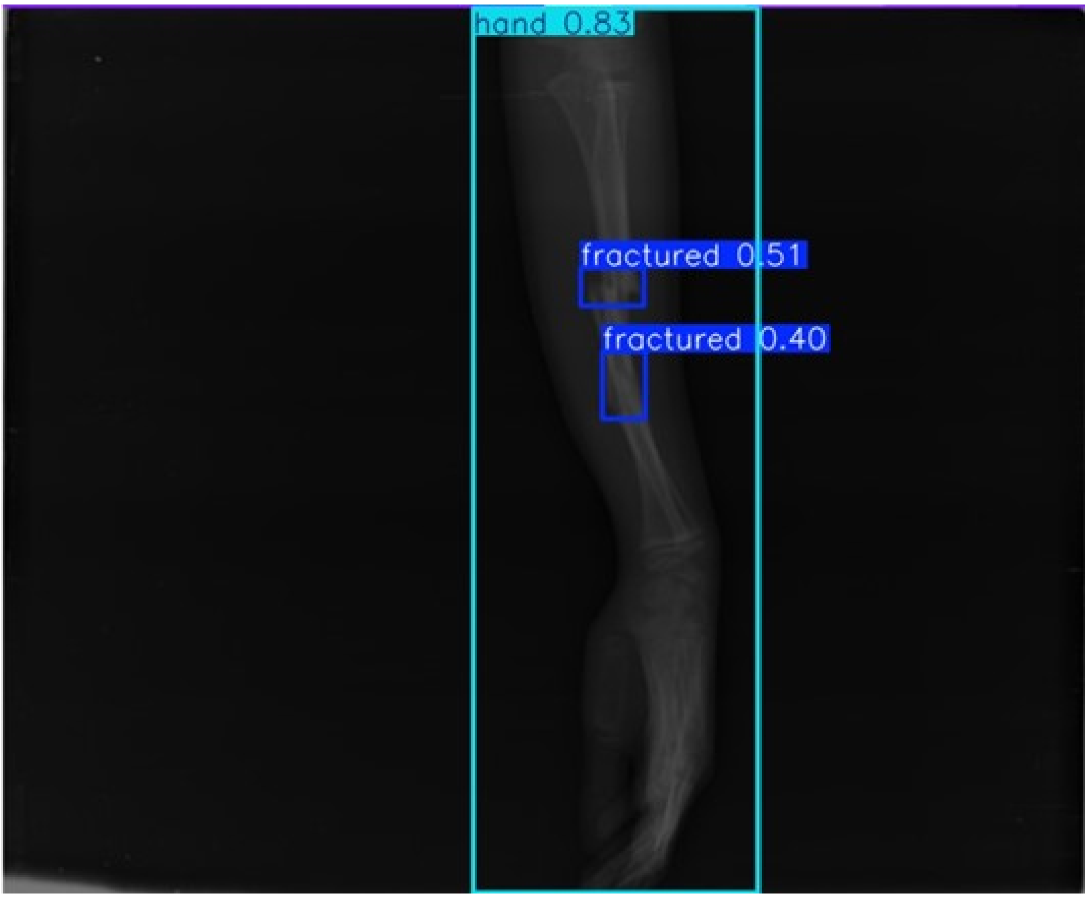


Figure 4.5: Sample output from the bone fracture detection model.

Class	Precision	Recall	F1-Score	Support
COVID19	1.00	1.00	1.00	214
NORMAL	0.94	0.97	0.95	557
PNEUMONIA	0.99	0.97	0.98	427
TUBERCULOSIS	0.96	0.93	0.95	351
Accuracy			**0.97**	
Macro Avg	**0.97**	**0.97**	**0.97**	**1549**
Weighted Avg	**0.97**	**0.97**	**0.97**	**1549**

Figure 4.6: Classification results of the chest disease detection model.

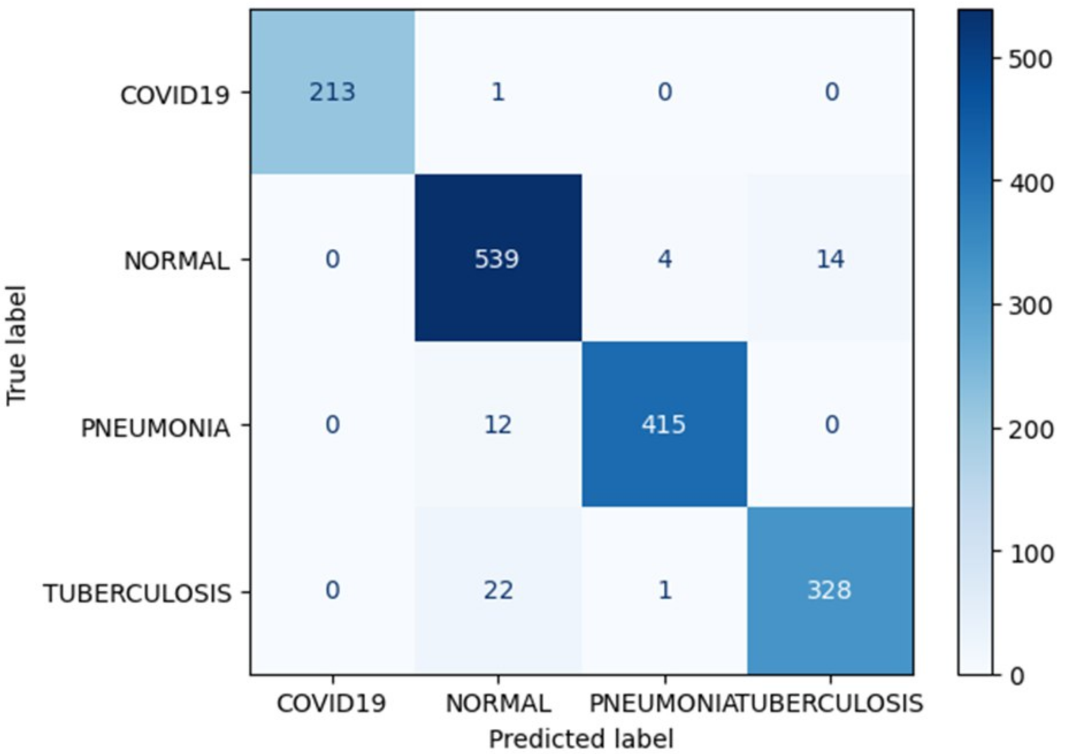


Figure 4.7: Confusion matrix of the chest disease detection model.

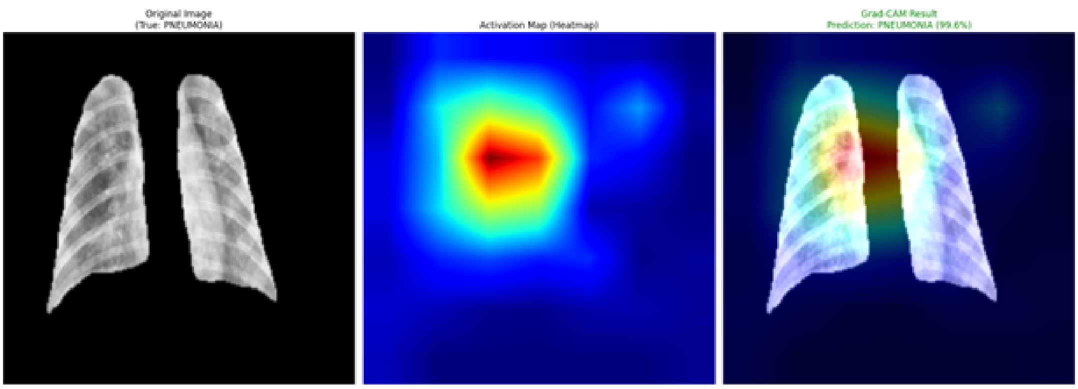


Figure 4.8: Sample output from the chest disease detection model.

The YOLOv8s had an input image size of 640×640 as raw data. It proved capable of finding normal bones, but had poor fracture recall.

In the next attempt, the resolution was increased to 1024×1024. The performance had a noticeable drop and the higher resolution introduced noise in the input data.

The following attempt, the high resolution was kept but Contrasted Limited Adaptive Histogram Equalization (CLAHE) was utilized to mitigate the problem of noise amplification. The performance recovered and the precision was high, but the recall remained a challenge.

Then, the newer YOLO11s model was tested on the MURA dataset to observe if it performed better than the previous attempts with the YOLOv8s model. It was observed that the performance of the YOLO11s model was worse than the YOLOv8 for this specific task.

In the final attempt, the batch size was configured to 8 to see if it had any significant effects on the model's performance. It was observed that reducing batch size allowed for more frequent weight updates, stabilizing gradients for the subtle fracture features. Thus, the final pipeline tested on the MURA dataset proved to be the best, and it employed the YOLOv8 model with CLAHE and a batch size of 8.

Figure 4.9 shows the results of the experiments done on the MURA dataset with the different YOLO models and configurations.

Model Config	Image Size	Precision	Recall	Fracture mAP50	Overall mAP50
YOLOv8s (Baseline)	640	0.744	0.410	0.553	0.683
YOLO11s	1024	0.712	0.375	0.444	0.587
YOLOv8m (Optimized)	**1024**	**0.648**	**0.467**	**0.572**	**0.694**

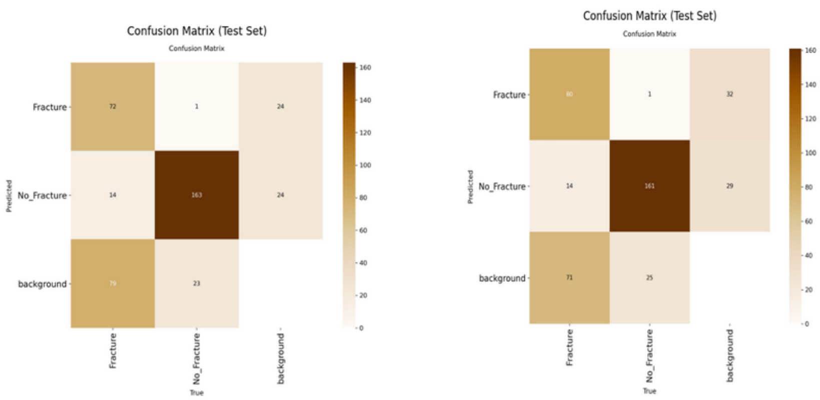


Figure 4.9: Results of successive experiments done on the MURA dataset to reach the best YOLO detection model possible.

References

[1] *AI-Based Applied Innovation for Fracture Detection in X-rays Using Custom CNN and Transfer Learning Models*. https://arxiv.org/pdf/2509.06228v1. 2025.

[2] "YOLOv11-based multi-task learning for enhanced bone fracture detection and classification in X-ray images". In: *ScienceDirect* (2025).

[3] *Bone Fracture Classification Using Transfer Learning*. https://arxiv.org/pdf/2406.15958. 2024.

[4] *Bone Fracture Detection and Classification using Deep Learning Approach*. https://www.researchgate.net/publication/341248451. 2020.

[5] "Classification of the Trabecular Bone Structure of Osteoporotic Patients Using Machine Vision". In: *ScienceDirect* (2017).

[6] *Bone Fracture Detection in X-ray Images using Convolutional Neural Network*. https://www.researchgate.net/publication/361433305. 2022.

[7] *Bone Fracture Detection from X-ray Images using a Convolutional Neural Network (CNN)*. https://www.researchgate.net/publication/384975763. 2024.

[8] A. Tahamtan and A. Ardebili. "Real-time RT-PCR in COVID-19 Detection: Issues Affecting the Results". In: *Expert Review of Molecular Diagnostics* 20.5 (2020), pp. 453–454.

[9] R. A. Boswell. "Pulmonary Tuberculosis: Clinical Features and Diagnosis". In: *Nurse Practitioner* 46.3 (2021), pp. 20–25.

[10] "Joint Diagnosis of Pneumonia, COVID-19, and Tuberculosis from Chest X-ray Images: A Deep Learning Approach". In: (2023).

[11] "Multi-class deep learning architecture for COVID-19, tuberculosis, and pneumonia classification". In: (2025).

[12] "Classification of COVID-19 from Tuberculosis and Pneumonia Using Deep Learning Techniques". In: (2022).

[13] "CXR-MultiTaskNet: Unified Deep Learning Framework for Disease Localization and Classification in Chest Radiographs". In: (2025).

[14] *Chest X-ray Classification Using Deep Learning for Automated COVID-19 Screening*. https://scispace.com/pdf/chest-x-ray-classification-using-deep-learning-for-automated-4jqra7h02h.pdf. 2020.

[15] "Detection of COVID-19, Pneumonia, and Tuberculosis from Radiographs Using AI-driven Knowledge Distillation". In: (2024).

[16] M.R. Bhuiyan and et al. Islam. "Deep learning-based analysis of COVID-19 X-ray images". In: *SAGE* (2023).

[17] "Joint Diagnosis of Pneumonia, COVID-19, and Tuberculosis from Chest X-ray Images". In: (2023).

[18] “Enhancing Wrist Fracture Detection and Classification through Deep Learning and XAI”. In: 2024.

[19] “Enhancing Bone Fracture Detection in Radiology: A Machine Learning and Explainable AI Approach”. In: 2024.

[20] “Interpretable Machine Learning for Osteoporosis Risk”. In: 2024.

[21] “XAI-based Computational Models for Predictive Analysis of Bone Fracture and Structural Repair”. In: 2024.

[22] *Explainable AI-Based Humerus Fracture Detection and Classification from X-Ray Images*. `https://www.researchgate.net/publication/389138164`. 2024.

[23] “Fracture Detection in Hand Radiographs with Grad-CAM Insights”. In: (2025).

[24] “An Explainable AI-Enabled Framework for Interpreting Pulmonary Diseases using Deep Learning”. In: (2023).

[25] “NextGen Lung Disease Diagnosis with Explainable Artificial Intelligence”. In: (2025).

[26] “Explainable AI in Deep Learning-Based Medical Image Analysis”. In: (2022).

[27] *Explainable Lung Disease Classification Using Deep Learning and XAI*. `https://arxiv.org/abs/2404.11428`. 2024.

[28] “Explanatory Classification of CXR Images into COVID-19, Pneumonia and Tuberculosis”. In: (2022).

[29] *Classifying Femur Fractures Using Federated Learning*. `https://www.diva-portal.org/smash/get/diva2:1831474/FULLTEXT01.pdf`. 2023.

[30] “A Systematic Review on Federated Learning in Medical Image Analysis”. In: (2023).

[31] “Privacy-Preserving Federated Learning for Collaborative Medical Data Mining”. In: (2025).

[32] “Boosting Multi-Demographic Federated Learning for Chest Radiograph Classification”. In: (2025).

[33] “Efficient Federated Learning for Pediatric Pneumonia on Chest X-ray Images”. In: (2024).

[34] *Federated Transfer Learning for Lung Disease Detection*. .

[35] “DMFL_Net: Federated Learning Framework for COVID-19 Classification”. In: (2023).

[36] “SemFedXAI: Semantic Framework for Explainable Federated Learning in Healthcare”. In: (2024).

[37] “FLEM-XAI: Federated Ensemble Model with Explainable AI for Lung Disease Diagnosis”. In: (2025).

[38] “FLPneXAINet: Federated Deep Learning and XAI for Pneumonia Detection”. In: (2025).

[39] “Secure and Interpretable Lung Cancer Prediction Using Blockchain Federated Learning with XAI”. In: (2025).